

Low-/negative-pressure hydrocephalus — literature review

Consolidated report over the 231-record corpus in `index.tsv`. Part I and Part III are computed from the index; Part II is the written synthesis from `notes/`. Pipeline: `harvest.py` → `curate.py` → `fetch_fulltext.py` → `gen_report.py` → `build_pdf.py`.

Part I — Corpus statistics (computed from `index.tsv`)

231 curated records, 1969–2026; 32 open-access full texts mirrored under `fulltext/` (`oa_xml`), 18 with a PMCID but no retrievable XML, 181 metadata-only.

Records by axis and evidence class

evidence class	Core entity	Mechanism	Treatment techniques	Aetiological settings	Adjacent differentials	Historical usage	all
systematic review	3	2	2		1		8
consensus / guideline	1						1
narrative review	6	7	1	1	4		19
case series	5		2	1	4	3	15
case series / cohort	19	16	21	9	14	10	89
case report	34	2	1	3	3	1	44
clinical / mechanistic study	3	3	1		2	12	21
imaging / physiology study		4					4
experimental (animal / bench)	3	8	2		2	1	16
computational model	1	8				1	10
comment / letter	4						4
all	79	50	30	14	30	28	231

Core-entity axis

79 records; 70 name the entity (low-/negative-/very-low-pressure hydrocephalus, SILPAH, ALPH) in title or abstract, 9 describe the phenomenon without the label.

Population: unspecified 46, adult 15, pediatric 13, mixed 5.

Precipitating setting tags (multi-label, core axis):

setting	n
shunt_dependent_prior	37
aqueductal_obstructive	17
infection_inflammatory	13
tumor_posterior_fossa	12
sah_aneurysm	11

setting	n
skull_base_csf_leak	11
lumbar_puncture_or_drain	9
trauma	6
craniectomy_skin_flap	6
intraventricular_hemorrhage	3

Treatment tags (multi-label, core axis):

treatment	n
shunt_or_valve	58
titrated_evd	35
subzero_evd	27
etv	12
csf_leak_repair	12
active_pump_shunt	10
elastography_imaging	8
neck_wrapping	7
icp_monitoring	5
positioning	3
pharmacologic	1

Publication years (core axis)

period	n
1970-1974	1
1985-1989	2
1990-1994	4
1995-1999	3
2000-2004	5
2005-2009	5
2010-2014	9
2015-2019	15
2020-2024	19
2025-2029	16

Part II — Synthesis

Sections below are the topical notes in `notes/` (00-08), concatenated. PMIDs cited resolve to rows of `index.tsv` except where a record is explicitly described as excluded from the corpus.

Low-/negative-pressure hydrocephalus — overview and executive summary

What the entity is

Low-pressure hydrocephalus (LPH), negative-pressure hydrocephalus (NegPH/NePH), very-low-pressure hydrocephalus (VLPH), acute low-pressure hydrocephalus (ALPH) and the "syndrome of inappropriately low-pressure acute hydrocephalus" (SILPAH) are names for one clinical paradox: **progressive ventriculomegaly with the symptoms of intracranial hypertension (depressed consciousness, headache, vomiting, cranial neuropathies, gait failure, bradycardia) while measured intracranial pressure is normal, low (conventionally ≤ 5 cm H₂O / ≤ 70 mm H₂O) or frankly sub-atmospheric** (down to -10 to -21 mm Hg in reported cases). Shunts, when present, are patent; conventional shunt revision or a standard EVD at 10–15 cm H₂O fails because the ventricular pressure never reaches the valve opening pressure or the drain outlet. Clinical status tracks ventricular size, not ICP (Pang & Altschuler 1994, PMID 7808607; Daniel 2002, PMID 12027920; Godoy Hurtado 2023, PMID 37373809).

The modern entity was defined by Pang & Altschuler (Neurosurgery 1994), who treated 12 shunted patients with ventriculomegaly and ICP 2.2–6.6 mm Hg by prolonged "sub-zero" external drainage and framed the mechanism as a viscoelastic alteration of brain parenchyma. Precursor descriptions exist from the 1970s–80s in shunted patients (McCullough & Fox 1974, PMID 4813717; McCullough 1986, PMID 3491341; Foltz & Blanks 1988, PMID 3343612). Before 1994, "low-pressure hydrocephalus" was used in Europe as a synonym for Adams-Hakim normal-pressure hydrocephalus; that literature is retained here as a separate axis so it is not conflated with the modern syndrome.

Corpus at a glance

- 231 curated records (1969–2026) from a 1,190-record Europe PMC harvest; 32 open-access full texts mirrored.
- **Core entity axis: 79 records**, 70 of which name the entity in the title/abstract; the remainder describe the phenomenon (symptomatic ventriculomegaly with patent shunt and low ICP) without the label.
- Evidence is almost entirely observational: 34 case reports, 24 case series / cohorts, 3 systematic reviews (Keough 2021 pooling 195 patients, PMID 32736355; Bakmeedeniya 2026 pooling 215 patients from 30 studies, PMID 41525875; Chang 2022 on Holmes tremor, PMID 35148507), 1 consensus statement (Chinese Neuroscience Society 2025, PMID 40734409) and no prospective comparative trial.
- Supporting axes: 50 brain-biomechanics / compliance / elastography papers, 30 drainage-and-valve technique papers (largely EVD-weaning and programmable-valve literature that frames titrated drainage), 14 etiological-setting papers, 30 adjacent differentials (slit-ventricle syndrome, overdrainage, intracranial hypotension, venous overdrainage), 28 historical LPH-as-NPH papers.

Ten take-home points

1. **Rare, under-recognised, and iatrogenically amplified.** Roughly 200–250 published cases exist (Godoy Hurtado 2023; Keough 2021; Bakmeedeniya 2026). Single-centre incidence after aneurysmal SAH is $\sim 3.7\%$ (Czorlich 2023, PMID 37004138); in neurosurgical patients with severe disorders of consciousness and ventriculomegaly, 60 % had low- or negative-pressure physiology (Zheng 2023, PMID 36994490). Almost every patient has had at least one prior intracranial procedure (mean 3.0 in Hunn 2014, PMID 24275338; ~ 3.5 operations per patient in Wu 2019, PMID 30611954).

2. **Precipitants cluster into three groups:** (a) a CSF sink outside the ventricles — lumbar puncture or lumbar drain, skull-base/cranial-wound CSF leak, pseudomeningocele, over-draining cyst or spinal shunt; (b) parenchymal "softening" — SAH/IVH, ventriculitis/meningitis, radiotherapy, trauma, diffuse injury, long-standing shunting; (c) loss of cranial containment — decompressive craniectomy / sinking-flap physiology.
3. **The mechanistic debate has four live models** — viscoelastic/hysteresis (Pang; Lesniak 2002, PMID 12617236), poroelastic "boggy brain"/Darcy flux (Akins 2011, PMID 21523524), transmantle gradient from ventricle-subarachnoid disconnection plus loss of brain turgor (Rekate 2008, PMID 18590387; Filippidis 2011, PMID 21800965; Diaz-Romero Paz 2019, PMID 30710718), and cerebral venous overdrainage/windkessel impedance (Barami 2016, PMID 27581321; Egnor 2023, PMID 37382303; Bateman 2026, PMID 42355921). They are not mutually exclusive and the leading reviews treat them as interacting (Bakmeedeniya 2026; Li 2024, PMID 38688300). MR elastography has shown reversibly low brain stiffness during an episode (Olivero 2016, PMID 27198914; Olivero 2021, PMID 33030603).
4. **Diagnosis rests on the paradox plus a therapeutic test:** symptomatic ventriculomegaly, ICP $\leq$ 5 cm H₂O on direct ventricular measurement, patent shunt / failed standard drainage, and improvement with drainage at zero or sub-atmospheric pressure (Pang-Altschuler criteria; Duan & Hu 2024, PMID 38579413; Datta & Tu 2025, PMID 41191101). Imaging shows effaced convexity sulci and Sylvian cisterns with disproportionate ventriculomegaly, often transependymal oedema despite low ICP, and — in the 2025 Chinese consensus — a fourth-ventricular "ball sign".
5. **Temporising treatment is titrated (usually sub-zero) EVD,** set to a target hourly output (5–15 mL/h, Hunn 2014) rather than a pressure, started at 0 to –5 cm H₂O below the EAM (Owler 2001, PMID 11599454), –3 to –10 cm (Akins 2011, PMID 21523524) or 5–10 cm below the midbrain (Foster 2016), lowered further only as needed (–15 in Godoy Hurtado 2023; –30 in one refractory case, Salam 2025), and then raised 1–3 cm H₂O every 3–5 days as ventricles shrink and ICP becomes positive (Akins 2011; Foster 2016, PMID 27069743). Drainage durations of weeks to months are typical (mean 41.9 days, range 2–365, in the pooled paediatric series of Chen 2025, PMID 40538927).
6. **Adjuncts that raise ICP or close the sink shorten or avoid sub-zero EVD:** neck wrapping / soft cervical collar (Filippidis 2011; Pandey 2017, PMID 28017745; Bruk 2026, PMID 42732439), Trendelenburg/recumbency (Foltz 1988; Dias 1999, PMID 10352418), epidural blood patch for post-LP or lumbar-drain cases (Dias 1999; Smalley 2017, PMID 28362957; Vilela 2020, PMID 32712403; Suzuki 2021, PMID 35854689; Hoffman 2026, PMID 42224725), repair of skull-base or wound CSF leaks (Filippidis 2011; Stevens 2024, PMID 39220674), removal of the offending spinal/cyst drain (Ozeki 2022, PMID 34526456; Diaz-Romero Paz 2019), and — a single striking case — theophylline (Salam 2025, PMID 40418891).
7. **ETV is the main strategy for reducing shunt dependence:** a "significant decrease" in shunt-dependent patients with ETV in SILPAH (10 vs 10 patients; Hamilton & Price 2012, PMID 22116443); 50 % ETV-only success in the Casado Pellejero algorithm (2022, PMID 34998487); ETV plus valveless catheter gave 81 % excellent outcomes in 16 adults, 19 % later needing a valved shunt (Villalonga 2025, PMID 41165379); but Foster 2016 (PMID 27069743) found no significant shortening of EVD time and in Chen 2025's pooled children 91 % ultimately needed shunt repositioning/revision.
8. **Definitive drainage needs a very-low-resistance conduit:** valveless VP shunts, programmable valves at the lowest setting without anti-siphon devices, "therapeutic siphoning" with non-siphon-control or adjustable gravitational valves (Bergsneider 1999, PMID 10520938; Trinh 2013, PMID 22770726; Funnell 2020, PMID 32989519), shunting into negative-pressure cavities (pleura, atrium), valves placed below the clavicle to exploit the hydrostatic column (Carrera 2025, PMID 39951714), wide-bore custom peritoneal catheters (Galbarriatu 2013, PMID 23904087), active-pumping shunts (Kalani 2013, PMID 23380444), improvised ultra-low valveless systems (Badran 2018, PMID 30287249) and intermittent reservoir pumping for chronic failure (Cheng 2017, PMID 28264586; Kuppler 2025, PMID 40030163).
9. **Outcome is bimodal and set by two things:** speed of recognition and the underlying disease. Pooled mortality is ~11 % (Keough 2021); 83 % of children but only 49 % of adults returned to their pre-episode baseline; 77.5 % of children recovered in Chen 2025; but Wu 2019's 39 secondary adult cases (18 with CNS infection) had a mean GOS-E of 2.7 and 8 deaths. Post-LP

paediatric cases are the most benign subgroup (Smalley 2017). Predictors of futility are unsealed CSF leak, irremediable ventriculitis and permanently altered compliance (Houlden 2018, PMID 30009119; Bakmeedeniya 2026).

10. **The evidence base is fragile:** heterogeneous definitions (ICP thresholds of 0, 5 and 7 cm H₂O; NegPH vs LPH; acute vs chronic), publication bias toward rescued cases, no registries, and mechanistic claims that rest on single MRE or phase-contrast measurements. Every major review from 2019 onward calls for standardised definitions and a multicentre registry.

How the notes are organised

- `01_history_nomenclature.md` — from 1969 "low-pressure hydrocephalus" (=NPH) through Pang & Altschuler to SILPAH, ALPH, cNegPH and pressure-based classification.
- `02_pathophysiology.md` — the competing/interacting models, experimental and imaging evidence, and what each model predicts about treatment.
- `03_etiologies_and_settings.md` — precipitants, frequencies in the pooled series, paediatric vs adult spectrum.
- `04_diagnosis.md` — phenotype, ICP measurement, imaging, differential diagnosis (overdrainage, slit ventricles, intracranial hypotension, NPH).
- `05_management.md` — acute algorithm (EVD titration, adjuncts, ETV), definitive CSF diversion hardware, chronic/refractory strategies, pharmacology.
- `06_outcomes_and_evidence_gaps.md` — outcome data, prognostic factors, evidence quality, and research questions.
- `07_key_papers.md` — per-paper extractions for the ~35 papers that carry the field.

History and nomenclature

1. "Low-pressure hydrocephalus" as a synonym for NPH (1965-c.1990)

After Adams, Hakim and colleagues described normal-pressure hydrocephalus in 1965, many European authors preferred the term low-pressure hydrocephalus for the same syndrome — an elderly patient with gait apraxia, dementia and incontinence, communicating ventriculomegaly and a lumbar opening pressure in the normal range. The 28 papers on the `historical_lph_as_nph` axis are of this kind: cerebral blood flow before and after shunting (Greitz 1969, PMID 5359210; Meixensberger 1989, PMID 2608784), isotope cisternography (Jensen 1970, PMID 5457870; Kulczycki 1978, PMID 683432), overnight pressure recordings that showed B-waves and plateau waves in "so-called low-pressure hydrocephalus" (Symon 1972, PMID 4117815; Sjaastad 1976, PMID 1278191), infusion tests (Bozzao 1974, PMID 4408072), psychometric profiles (Rivero Hernández 1975, PMID 1211270) and shunt series with "disappointing results" (Bannister 1972, PMID 4540544; Singounas 1976, PMID 1250499). Symon's demonstration that these patients have intermittently raised pressure is a reason the term fell out of use for NPH.

These papers are relevant to the modern entity in two ways only: they show that ventricular enlargement can progress without sustained hypertension, and they are the source of persistent terminological confusion (e.g. a 2013 case report still describing "acute normal pressure hydrocephalus" for what is clearly post-LP low-pressure disease; Preuß 2013, PMID 23715811). The Kuba 2002 gait study (PMID 11749014) and several 1970s "LPH" reports were retained in the core axis because their abstracts describe low measured pressure with progressive ventriculomegaly, but they should be read as NPH-era work.

2. Precursors in shunted patients (1974-1993)

- **McCullough & Fox 1974** (PMID 4813717): negative ICP in shunted adults and its relation to subdural haematoma — the first systematic recognition that shunted brains can run at sub-atmospheric pressure, here framed as an overdrainage complication rather than a hydrocephalic state.
- **McCullough 1986** (PMID 3491341): symptomatic progressive ventriculomegaly in hydrocephalics with patent shunts and antisiphon devices — the essential paradox (large ventricles, working shunt, low pressure) stated a decade before it was named. The anti-siphon device prevented the shunt from ever reaching the sub-zero pressures the brain needed.
- **Foltz & Blanks 1988** (PMID 3343612): symptomatic low intracranial pressure in shunted hydrocephalus — upright, activity-related symptoms with markedly low upright ICP, relieved by recumbency and Trendelenburg; Foltz later designed a "zero-ICP" gravity-controlled shunt (ZIPS; Foltz 1994, PMID 8194062).
- **Drake 1993** (PMID 8421544): functional obstruction of an anti-siphon device by tissue-capsule pressure — an early mechanical explanation for "patent shunt, large ventricles".
- **Pomeranz 1993** (PMID 8333302): feline intracranial hypotension reduces cerebral blood flow — an experimental hint that low ICP itself can be harmful.

3. Pang & Altschuler 1994 and the definitional decade

Pang & Altschuler, Neurosurgery 1994 (PMID 7808607) coined low-pressure hydrocephalic state: 12 shunted patients with symptomatic ventriculomegaly, ICP 2.2-6.6 mm Hg, patent shunts, requiring prolonged EVD at sub-zero levels (most for weeks), with symptoms tracking ventricular size rather than ICP. They proposed altered brain viscoelasticity — a "softened" parenchyma that no longer returns to shape under a normal transmante gradient — and established sub-zero drainage as the reference treatment. Their four criteria (symptoms of hydrocephalus, ventriculomegaly, ICP $\leq$ 5 cm H₂O, improvement with sub-zero drainage) are still used for inclusion (Foster 2016; Villalonga 2025).

Subsequent naming:

Term	Source	Nuance
Negative-pressure hydrocephalus (NegPH, NePH)	Vassilyadi 1995 (PMID 7666227), after a spinal arachnoid cyst-pleural shunt	ICP actually sub-atmospheric; implies a transmante gradient created by an external sink
Low-pressure shunt "malfunction" after LP	Dias 1999 (PMID 10352418)	Paediatric, post-lumbar-puncture; resolves with recumbency $\pm$ blood patch
Very-low-pressure hydrocephalus (VLPH)	Clarke 2006 (PMID 16961147); Houlden 2018 (PMID 30009119)	Refractory subgroup with poor outcome after suprasellar/skull-base surgery
Hysteresis / non-linear dynamics	Lesniak 2002 (PMID 12617236)	Mechanistic label; 10 adults, mean ICP 4.8 mm Hg
Cortical subarachnoid space / brain turgor model	Rekate 2008 (PMID 18590387)	Reframes as ventricle-SAS disconnection; "compliance" replaced by "turgor"
Syndrome of inappropriately low-pressure acute hydrocephalus (SILPAH)	Hamilton & Price 2012 (PMID 22116443)	Emphasises acute deterioration and ETV as treatment
Acute low-pressure hydrocephalus (ALPH)	Keough 2021 (PMID 32736355); Czorlich 2023 (PMID 37004138)	Systematic-review label; ICP $\leq$ 5 cm H ₂ O typical
Chronic negative-pressure hydrocephalus (cNegPH)	Kuppler 2025 (PMID 40030163)	Persistent ventriculomegaly after failure of prolonged sub-atmospheric drainage
Ultra-low-pressure hydrocephalic state in NPH	Funnell 2020 (PMID 32989519)	Elderly iNPH patients who relapse after shunting and need siphoning
Two-compartment / acute "normal pressure" hydrocephalus	Preuß 2013 (PMID 23715811)	Idiosyncratic label for post-LP LPH in a shunted child

Term	Source	Nuance
Low-/negative-pressure hydrocephalus (LPH/NePH, L&NPH)	Wu 2019; Li 2024; Chen 2025	Chinese-language literature; LPH 0-70 mm H ₂ O, NegPH ≤ 0

4. Classification proposals

- **Wu 2019** (PMID 30611954) proposed classifying hydrocephalus by ventricular pressure — high, normal, low (< 70 mm H₂O) and negative (< 0) — arguing that existing schemes (Oi's multi-categorical classification, PMID 21928018; Raimondi's unifying theory, PMID 8194058; Rekate's 2008 definition based on points of obstruction) give no management guidance for the low-pressure group.
- **Chinese Neuroscience Society 2025 consensus** (PMID 40734409) defines LPH as persistent intraventricular pressure < 70 mm H₂O with progressive ventriculomegaly and neurological worsening, adds imaging criteria (effaced Sylvian cisterns and convexity SAS, fourth-ventricular "ball sign"), and names fourth-ventricular outlet obstruction and altered compliance as candidate mechanisms. It is the first formal guideline document on the entity.
- **Lin 2025** (PMID 40826122) places low-pressure hydrocephalus as the second most common subtype of "refractory hydrocephalus" after infectious hydrocephalus and stresses that management hinges on "identifying and addressing the sites receiving CSF".
- **Rekate 2019** "Low or Negative Pressure Hydrocephalus Demystified" (PMID 31103762) and **Rekate 2021** commentary (PMID 34272587) restate the turgor/disconnection view and argue the syndrome is a predictable consequence of CSF pathway anatomy rather than a mystery.

5. Terminological problems that remain

1. **Threshold:** ≤ 5 cm H₂O (Pang, Keough, Datta), < 70 mm H₂O (Wu, Chinese consensus), ≤ 0 (NegPH) and "below valve opening pressure" (Filippidis) are all in use; a case at 6–8 cm H₂O with a patent shunt set at 10 is LPH to some authors and shunt under-drainage to others.
2. **Acute vs chronic:** ALPH/SILPAH (deterioration over hours-days after a precipitant) and cNegPH / ultra-low-pressure NPH (months of refractory ventriculomegaly) probably share physiology but differ in urgency and in the role of ETV.
3. **NPH overlap:** Funnell 2020 and Trinh 2013 describe shunted iNPH patients who need siphoning; Doshi 2022 (PMID 36600762) describes an NPH patient evolving into VLP. Whether "low-pressure NPH" is the same disease as post-SAH ALPH is unsettled.
4. **Sinking-flap / paradoxical herniation overlap:** after craniectomy the same sub-atmospheric physiology produces sinking flap syndrome, paradoxical herniation and LPH, sometimes sequentially in one patient (Bateman 2026, PMID 42355921; Salam 2025).

Pathophysiology — competing and interacting models

The central puzzle: ventricles enlarge and compress brain even though the pressure inside them is not high. Every model must therefore explain either (i) a gradient across the brain mantle that exists despite a low absolute ventricular pressure, or (ii) a change in the material of the brain such that a normal or low pressure is enough to keep it distended, or both. The systematic review of Bakmeedeniya 2026 (PMID 41525875) identified four commonly invoked models — viscoelastic change, transmantle gradients, poroelastic/glymphatic dysfunction, impaired CSF pulsatility — and treats them as interacting. The clinician-oriented review of Li 2024 (PMID 38688300) covers the same four under the labels transcortical pressure difference, excessive cerebral venous drainage, brain viscoelastic change and the "porous elastic sponge".

1. Viscoelastic softening and hysteresis (Pang & Altschuler; Lesniak)

- **Claim.** Chronic or intermittent distension (e.g. repeated shunt failures with high pressure) permanently alters the stress-strain behaviour of periventricular white matter. Hooke's-law elasticity is replaced by viscous creep: the brain behaves like a material that has been stretched past its yield point and does not recoil when pressure normalises. A lower-than-normal ventricular pressure is therefore required to restore shape (Pang & Altschuler 1994, PMID 7808607).
- **Hysteresis.** Lesniak 2002 (PMID 12617236) modelled 10 adults (mean ICP 4.8 mm Hg) as a pressure-volume loop with hysteresis: the volume reached at a given pressure depends on the path taken. Only negative-pressure drainage moved the system back onto the low-volume limb; ventricles then stayed small at normal pressure. Godoy Hurtado 2023 (PMID 37373809) uses the lung analogy (inflation and deflation limbs differ).
- **Evidence.** MR elastography in a 19-year-old with shunted aqueductal stenosis showed very low shear stiffness during an LPH episode (Olivero 2016, PMID 27198914) that recovered to near-normal two years later after a high-pressure shunt failure (Olivero 2021, PMID 33030603) — the only direct human measurement of reversibly reduced stiffness in LPH. Houlden 2018 (PMID 30009119) argued that recovery of GCS despite fluctuating frontal-horn ratio indicates "retained viscoelasticity" and predicts salvageability. Supporting biomechanics: brain elasticity changes with ventriculomegaly (Sklar 1980, PMID 6893601), reassessment of brain elasticity in hydrocephalus models (Taylor & Miller 2004, PMID 15212932), MRE stiffness falls in early-onset hydrocephalus white matter (Wagshul 2021, PMID 33631603) and rises in NPH with partial normalisation after shunting (Streitberger 2011, PMID 20931563; Freimann 2012, PMID 21538046; Fattahi 2016, PMID 26542235).
- **Weakness.** Does not explain why the syndrome is so often triggered by an acute external CSF sink (LP, leak) in patients whose brains were not chronically distended, or why neck wrapping works within hours.

2. Poroelastic "boggy brain" — Darcy flux of interstitial fluid (Akins)

- **Claim.** The brain is a fluid-saturated porous solid. When interstitial fluid accumulates (transependymal oedema, inflammation, radiation injury), the parenchyma loses turgor and its ability to resist deformation; ventricles enlarge at low pressure. Sub-zero drainage produces a transient "CSF diuresis" (drain output exceeds baseline for days) as interstitial water is pulled back into the ventricles along a Darcy flux, after which the brain re-stiffens (Akins 2011, PMID 21523524: 9 patients, transependymal oedema in 5 despite low ICP, EVD output fell as LPH developed and rose above baseline during sub-zero drainage).
- **Model support.** Poroelastic finite-element models of hydrocephalus (Peña 1999, PMID 10414573; Dutta-Roy 2008, PMID 18534602; Chou 2016, PMID 26671218; Kim 2016 on periventricular lucency, PMID 26274984) show that small sustained gradients can produce large ventricular strain when permeability and elasticity are altered; Solamen 2021 (PMID 32331966) measured poroelastic properties of NPH brains during lumbar drainage. Duy 2024 (PMID 38798141) reviews hydrocephalus as "biomechanical instability of the brain-CSF interface".
- **Clinical corollary.** Akins' group used the "sub-zero method" successfully in all 9 patients and internalised with low-pressure hardware; the model predicts that fluid shifts (hyponatraemia, low osmolarity) worsen LPH — consistent with Houlden 2018's observation that avoiding fluid shifts helped, and with Pandey 2017 (PMID 28017745) deliberately lowering plasma osmolarity to raise ICP.

3. Transmantle gradient from ventricle-subarachnoid disconnection and loss of turgor (Rekate/Filippidis)

- **Claim.** Ventricular size is set by the pressure difference between the ventricle and the cortical subarachnoid space (CSAS), not by absolute ICP. If the CSAS is (a) disconnected from the ventricles by obstruction at the aqueduct, fourth-ventricular outlets, basal cisterns or tentorial incisura, and (b) drained to a lower pressure than the ventricles by an external sink (lumbar puncture/drain, skull-base leak, cyst-pleural shunt), then the ventricle is at higher pressure than

the SAS even when both are sub-atmospheric; CSF cannot return to the convexities and the ventricles enlarge (Rekate 2008, PMID 18590387; Filippidis 2011, PMID 21800965; Vassilyadi 1995, PMID 7666227; Diaz-Romero Paz 2019, PMID 30710718; Suzuki 2021, PMID 35854689). Rekate replaces "compliance" with **brain turgor** — the capillary-bed-dependent firmness that normally keeps ventricles small — which is reduced by ischaemia, radiotherapy, injury and low venous pressure.

- **Predictions that have held.** (i) Neck wrapping / jugular compression raises venous and hence CSAS pressure and turgor and reverses ventriculomegaly within hours to days (Filippidis 2011; Pandey 2017; Bruk 2026, PMID 42732439); healthy-volunteer MRE confirms jugular compression stiffens brain in proportion to jugular venous flow (Hatt 2015, PMID 26045579). (ii) Closing the sink (blood patch, leak repair, removing a spinal drain) can abort the syndrome without sub-zero EVD (Dias 1999; Vilela 2020; Suzuki 2021; Ozeki 2022; Hoffman 2026). (iii) ETV, which reconnects ventricle and SAS, reduces shunt dependence (Hamilton & Price 2012). (iv) Lumbar-ventricular pressure discrepancies are measurable in obstructed systems (Chang 2026, PMID 42218510; Staykov 2011, PMID 21273925).
- **Counter-evidence and nuance.** Direct measurements have failed to find a transmante gradient in communicating or non-communicating hydrocephalus at steady state (Stephensen 2002, PMID 11904027; Shapiro 1987 cats, PMID 3598675; Eide 2010 on pulsatile gradient, PMID 20130957), while Conner 1984 (PMID 6737056) did find one in experimental NPH. Advocates reply that the relevant gradient is small, dynamic and only present while a sink is active, and that MRI-derived transmante pressures (Sincomb 2021, PMID 34385144) are on the order of Pascals — enough over time in a soft brain.

4. Cerebral venous overdrainage, windkessel and pulsatility models

- **Venous overdrainage (Barami 2016, PMID 27581321).** CSF diversion at low pressure lowers the pressure surrounding cortical veins and bridging veins, collapsing them or increasing venous outflow, reducing cerebral blood volume and turgor; the brain "deflates" toward the ventricles. Vilela 2020 (PMID 32712403) invoked this for post-LP LPH in an adult; Datta & Tu 2025 (PMID 41191101) found prior jugular/sigmoid sinus thrombosis in a third of their paediatric LPH patients and modelled venous thrombosis as a predisposing factor; Sainz 2019 (PMID 30474714) showed the mirror image — venous hypertension increases turgor and prevents ventriculomegaly in infant "external hydrocephalus".
- **Windkessel / CSF path impedance (Egnor 2023, PMID 37382303; Egnor 2026, PMID 41791099).** In a lumped electrical model the cranium is a band-stop filter that shunts arterial pulsatile energy through CSF to the veins. Egnor classifies hydrocephalus by CSF-path impedance: obstructive = high resistance; NPH = low inertance and high compliance; **low-pressure hydrocephalus = high resistance and high compliance**. Ventriculomegaly is an adaptive response that lowers impedance, which is why it can occur at any absolute pressure. Related pulse-wave-encephalopathy work (Bateman 2002, PMID 12221445; Bateman 2008, PMID 18379767) and pulsatility-driven glymphatic flow (Nozalea 2025, PMID 40258946) feed this view.
- **Autoregulatory reset and capillary CSF absorption (Bateman 2026, PMID 42355921).** In a craniectomy patient who passed through sinking-flap syndrome and LPH, phase-contrast MRI showed bilaterally reduced CBF that rose by 26 % after cranioplasty and reversed aqueductal CSF flow toward the ventricles during the LPH phase. Lumped-parameter modelling favoured arteriolar constriction from an autoregulation reset over venous compression, and the authors hypothesise increased CSF absorption via the capillary bed at low ICP.
- **Unified mathematical model (Wang & Lu 2020 preprint).** A U-shaped ICP-volume curve that claims to explain slit ventricles, NPH/LPH and paradoxical herniation in one framework; untested.

5. Compliance, elastance and pressure-volume physiology (supporting axis)

The 50 biomechanics papers give the vocabulary and measurements the LPH literature borrows:

- **Pressure-volume compensation** (Czosnyka 1996, PMID 8778261; Czosnyka 2001 age dependence, PMID 11235954; Kim 2009 compensatory reserve index, PMID 19240611; Gholampour 2022 new definition of compliance after shunting, PMID 35979170). LPH is a state of high compliance / low elastance: large volume change for little pressure change — the opposite of slit-ventricle syndrome.
- **Age and species dependence of viscoelasticity** (Shulyakov 2011, PMID 21316637; Shulyakov 2012 acute experimental hydrocephalus, PMID 22806081; Jugé 2016 rat obstructive hydrocephalus microstructure and stiffness, PMID 26848844; Pattison 2010 feline MRE, PMID 20655045; Dirrichs 2019 infant shear-wave elastography, PMID 31464808). Infant brains are uniquely compliant — Rekate 2020 (PMID 32488353) argues this is why infant hydrocephalus, sinking fontanelles and post-LP LPH behave differently.
- **MR elastography in NPH and dementia** (Perry 2017, PMID 28063896; ElSheikh 2017, PMID 28570101; Murphy 2020, PMID 32058331; Aunan-Diop 2022 scoping review, PMID 34687356; Coelho 2022 systematic review, PMID 35488708; Karki 2024 outcome prediction, PMID 38272572; Hersh 2022 operative applications, PMID 35933785). Stiffness increases periventricularly in NPH and falls after shunting — the opposite direction from the LPH case reports, suggesting LPH is not simply an extreme NPH.
- **Historical theory** — Raimondi's unifying definition (PMID 8194058), Orešković & Klarica's critique of the bulk-flow hypothesis (PMID 21641963; choroid plexus role, PMID 28456718), Oi's evolution theory of CSF dynamics (PMID 16685545), Li 2008 kaolin models of cortical SAS obstruction (PMID 18433747), Fukuhara 2001 shunt removal and compliance (PMID 11302655).

6. Synthesis: a two-hit, two-compartment picture

The most parsimonious reading of the corpus is that LPH requires **a soft or low-turgor brain** (hit 1: haemorrhage, infection, radiation, chronic distension, craniectomy, venous thrombosis, immaturity) **and a pressure differential that favours the ventricle** (hit 2: an external CSF sink or a disconnected SAS at lower pressure, or shunt hardware that cannot drain below its opening pressure). Treatments map onto the two hits:

Intervention	Acts on
Sub-zero EVD	Makes the ventricle the lowest-pressure compartment; unloads interstitial water (hit 2, then 1)
Neck wrapping / collar / Trendelenburg / hypercapnia / low osmolarity	Raises venous pressure, CBV and turgor (hit 1)
Blood patch, leak repair, removing spinal/cyst drain	Removes the sink (hit 2)
ETV	Reconnects ventricle and SAS so the gradient cannot form (hit 2)
Valveless / siphoning shunts, negative-pressure cavities, active pumps	Permanent low-resistance path so the ventricle never sits above the SAS (hit 2)
Theophylline	Raises ICP in intracranial hypotension — mechanism uncertain, possibly adenosine antagonism and vasoconstriction restoring CSF volume/turgor (hit 1)

The field has not tested these models against each other; the only human material-property data are two MRE scans in one patient.

Etiologies, precipitants and clinical settings

1. What the pooled series say

Series	n	Population	Underlying hydrocephalus etiology	Precipitant / context
Pang & Altschuler 1994 (PMID 7808607)	12	mostly adults	shunted hydrocephalus of varied cause	shunt malfunction / prolonged shunting
Owler 2001 (PMID 11599454)	5	mixed	tumour, SAH, prior shunts	postoperative
Lesniak 2002 (PMID 12617236)	10	adults	mixed	repeated shunt failures
Akins 2011 (PMID 21523524)	9	adults (neuro-ICU)	SAH, chronic hydrocephalus, tumour, chronic CNS infection	4/9 during index admission for intracranial bleeding; 7 days-5 yrs after primary dx
Hamilton & Price 2012 (PMID 22116443)	20	children + adults	meningitis, post-LP, intraventricular tumours	acute deterioration
Hunn 2014 (PMID 24275338)	8	adults	mixed	all had ≥ 1 prior intracranial procedure (mean 3.0)
Foster 2016 (PMID 27069743)	16	adults	mixed	prolonged EVD
Smalley 2017 (PMID 28362957)	29 (30 episodes)	children	tumour, haemorrhage	11/30 after lumbar puncture; 19 spontaneous
Cheng 2017 (PMID 28264586)	7	adults	—	VP shunt at lowest setting failing
Wu 2019 (PMID 30611954)	39	adults	TBI, haemorrhage, tumour resection, CNS infection (18/39)	~3.5 operations per patient
Keough 2021 systematic review (PMID 32736355)	195 (42 local + 153 literature)	53 children, 142 adults	neoplasm and haemorrhage most common	39 % of children post-LP; shunt in situ in 92 % of children and 39 % of adults
Casado Pellejero 2022 (PMID 34998487)	5	adults	mixed	prior shunt / infection / surgery
Czorlich 2023 (PMID 37004138)	15 of 406 SAH (3.7 %)	adults	aneurysmal SAH	IVH presence and extent; mean onset day 13
Godoy Hurtado 2023 (PMID 37373809)	6	mixed	medulloblastoma $\times 2$, TBI with bifrontal craniectomy, craniopharyngioma, leptomeningeal glioneuronal tumour, shunted NPH	4/6 already had mid/low-pressure shunts
Zheng 2023 (PMID 36994490)	41 of 68 DOC patients with hydrocephalus (60 %)	adults with severe disorders of consciousness	TBI/haemorrhage with craniectomy	ventriculomegaly + poor compliance

Series	n	Population	Underlying hydrocephalus etiology	Precipitant / context
Chen 2025 pooled paediatric (PMID 40538927)	48 (44 literature + 4 own)	children, mean age 6–8.5 y	neoplasm 71 % (medulloblastoma, astrocytoma 59 % of tumours), haemorrhage 15 % (incl. PHH of prematurity), TBI 8 %, congenital (Dandy- Walker, craniosynostosis, aqueduct stenosis, NF1)	> 50 % had prior surgery; 41 % "spontaneous" after primary treatment
Datta & Tu 2025 (PMID 41191101)	6	children	—	prior jugular/sigmoid sinus thrombosis in 2/6
Villalonga 2025 (PMID 41165379)	16	adults	—	Pang-Altschuler criteria
Bakmeedeniya 2026 systematic review (PMID 41525875)	215 from 30 studies	mixed	TBI, aneurysmal SAH, NPH, postoperative, spontaneous	—

2. Precipitants by mechanism class

A. An external or extra-ventricular CSF sink

- **Lumbar puncture** in a shunted child or adult with obstructive hydrocephalus (Dias 1999, PMID 10352418; Preuß 2013, PMID 23715811; Smalley 2017; Vilela 2020, PMID 32712403; Koenig 2025, PMID 41357016; Hoffman 2026, PMID 42224725). 37–39 % of paediatric cases are post-LP (Smalley; Keough). The shunt is patent; the LP creates a spinal SAS at lower pressure than the ventricle across an obstructed fourth ventricle/aqueduct.
- **Lumbar or spinal drainage** after SAH or posterior-fossa surgery (Ozeki 2022, PMID 34526456; Suzuki 2021, PMID 35854689; Dagnew 2002 acquired Chiari after lumbar drainage, PMID 12188966; Staykov 2011 lumboventricular gradient, PMID 21273925; Bakhshi 2020 lumbar-drain complications, PMID 32711138).
- **Skull-base and cranial-wound CSF leaks / pseudomeningocele** (Filippidis 2011, PMID 21800965 — 3 cases after cranial-base approaches, all with documented or presumed ventricle-SAS block; Stevens 2024 skull-base leak with tension pneumocephalus, PMID 39220674; Smith 2016 posterior fossa pseudomeningocele, PMID 29506116; Udayakumaran 2018 leaks after tethered-cord surgery, PMID 29605699; Bakhsh 2022 LP-shunt tract leak, PMID 35673636; Kumar 2025 extra-axial ETV for persistent leak, PMID 41215506).
- **Over-draining cyst or spinal shunts** (Vassilyadi 1995 spinal arachnoid cyst-pleural shunt, PMID 7666227; Diaz-Romero Paz 2019 external cyst drain, PMID 30710718; Marianayagam 2017 trapped fourth ventricle treatment, PMID 28209310).
- **Hemispherectomy with CSF rhinorrhoea** (Daniel 2002, PMID 12027920).

B. Parenchymal softening / loss of turgor

- **Aneurysmal SAH and IVH** — 3.7 % of SAH patients; IVH extent is the risk factor; aLPH patients later needed permanent shunts far more often (66.7 % vs 17.4 %) (Czorlich 2023). Case reports: Wang 2024 with quantitative EEG warning (PMID 39740216); Weisbrod 2021 angiogram-negative SAH after COVID-19 (PMID 34462697); Badran 2018 haemangioblastomatosis SAH (PMID 30287249). Background: Chen 2017 and Kuo 2021 reviews of post-SAH hydrocephalus (PMIDs 28373987, 34068783).
- **Ventriculitis / meningitis / chronic CNS infection** — 18/39 in Wu 2019; Akins 2011; Salam 2025; Kuppler 2025. Infection is also the strongest predictor of poor outcome (Bakmeedeniya 2026; Houlden 2018).

- **Brain tumour and its treatment** — posterior-fossa tumours in children (medulloblastoma, ependymoma, astrocytoma), suprasellar lesions (craniopharyngioma; Houlden 2018's four VLP cases followed suprasellar resection), tectal tumours (Chang 2022 Holmes tremor, PMID 35148507), leptomeningeal/intraventricular spread (Godoy 2023; Myers 2023), radiotherapy (Duan & Hu 2024 list prior cranial radiotherapy as a risk factor).
- **Traumatic brain injury** — Wu 2019; Svedung Wettervik 2022 (PMID 34414834: patients with "low-pressure hydrocephalus symptoms" had the best shunt response among post-traumatic hydrocephalus patients); Salam 2025; He 2026 (PMID 42444806).
- **Long-standing shunt dependence** — 81 of the 231 records carry the `shunt_dependent_prior` tag; Pang's original series was entirely shunted patients; Keough found a shunt in situ in 92 % of children.
- **Sepsis / cardiovascular collapse** (McGovern 2013, PMID 23816322) and **COVID-19** (Weisbrod 2021) — single cases suggesting systemic inflammatory/vascular insults can precipitate the state.

C. Loss of cranial containment

- **Decompressive craniectomy / sinking-flap physiology** — Godoy 2023 case 3; Salam 2025; Bateman 2026 (PMID 42355921); Bhatjiwale 2023 malignant oedema after cranioplasty (PMID 37680924); Ding 2014 review of craniectomy and hydrocephalus (PMID 25252237); Kutty 2018 (PMID 29725175) and Wan 2016 (PMID 26971294) on managing ventriculomegaly and abnormal flap pressure; Joshua 2011 papilloedema after craniectomy (PMID 21676532); Zheng 2023 combined cranioplasty + shunt in DOC patients. The open cranium abolishes the rigid container assumed by Monro-Kellie, so atmospheric pressure acts directly on the brain and small gradients translate into large volume shifts.

D. Shunt hardware as precipitant

- Anti-siphon devices and gravitational units that prevent sub-zero drainage in a brain that needs it (McCullough 1986; Drake 1993; Trinh 2013; Funnell 2020).
- Programmable valves left at high settings after over-drainage complications (Marianayagam 2017; Doshi 2022 with raised intra-abdominal pressure, PMID 36600762).
- Ventriculopleural/atrial shunts and the general risk of negative-pressure cavities — Chiang 2001 (excluded from core), Datta & Tu 2025 (used deliberately).

3. Paediatric versus adult spectrum

Feature	Children	Adults
Dominant etiology	Posterior-fossa tumour (71 % neoplasm in Chen 2025), PHH of prematurity, congenital malformations	SAH/IVH, TBI, CNS infection, postoperative skull-base leak, shunted NPH
Precipitant	Lumbar puncture in 37–39 %	Usually spontaneous after surgery, drain or infection
Shunt in situ at onset	92 % (Keough)	39 % (Keough)
Distinctive symptoms	Bradycardia, seizures, failure to thrive, vomiting (Chen 2025; Smalley 2017)	Depressed consciousness, headache, gait failure, cranial nerve palsies
Course of post-LP cases	Often benign: recumbency $\pm$ blood patch; median 4 days of external drainage vs 11 for spontaneous (Smalley)	Blood patch effective in adults too (Vilela 2020)
ETV	Rarely sufficient alone; 91 % need shunt repositioning (Chen 2025)	Reduces shunt dependence (Hamilton); ETV-only success 50 % (Casado Pellejero)
Return to baseline	83 % (Keough); 77.5 % (Chen)	49 % (Keough); poor in infected/DOC cohorts (Wu 2019)

Infant brain compliance (Rekate 2020, PMID 32488353; Welch 1980, PMID 7373397; Dirrichs 2019) and the frequency of obstructive posterior-fossa pathology explain why the paediatric phenotype is dominated by the LP-sink mechanism and why it recovers more completely.

4. Frequency estimates

- Post-SAH: 15/406 = 3.7 % (Czorlich 2023).
- Severe DOC with hydrocephalus: 41/68 = 60 % low or negative pressure (Zheng 2023) — the highest prevalence reported, suggesting that in this population LPH is the rule and that recovery of consciousness may depend on recognising it.
- Post-traumatic hydrocephalus: 14/29 shunted patients had "impeded recovery or low-pressure hydrocephalus symptoms" (Svedung Wettervik 2022).
- Titanium-mesh exposure after cranioplasty with a shunt: 9/10 had LPH (Zheng 2025, PMID 39760505; excluded from core but illustrates the association with chronic intracranial hypotension).
- Cumulative published cases: ~90 (Strand 2018, 25-year search), ~195 (Keough 2021), ~200 (Godoy 2023), 215 (Bakmeedeniya 2026). The Chinese consensus states incidence is "systematically underestimated" and no epidemiology exists.

Diagnosis, imaging, ICP findings and differential diagnosis

1. Clinical phenotype

Symptoms are those of acute or subacute hydrocephalus and are indistinguishable at the bedside from high-pressure disease:

- Depressed level of consciousness — the predominant sign in the pooled 195 (Keough 2021, PMID 32736355); may progress to coma with brain-stem signs (Hunn 2014 mean GCS rise 4.6 on treatment, PMID 24275338).
- Headache, nausea/vomiting (nausea/vomiting in half of Smalley's children), lethargy, confusion.
- Cranial neuropathies (up-gaze palsy, Parinaud's syndrome with tectal compression), ataxia and gait failure, urinary retention/incontinence.
- Paediatric-specific: bradycardia, seizures, failure to thrive (Chen 2025, PMID 40538927).
- Movement disorder: Holmes tremor with triventricular dilation (Chang 2022, PMID 35148507).
- **Postural pattern:** symptoms worse upright and relieved recumbent in the Foltz/Dias post-LP phenotype (Foltz 1988, PMID 3343612; Dias 1999, PMID 10352418; Ozeki 2022 headache on head elevation, PMID 34526456) — the same orthostatic pattern as intracranial hypotension, which is a clue that the two entities share physiology.
- **Timing:** after a precipitant (LP, drain, leak) deterioration occurs within hours to days; after SAH the mean latency was 13.1 ± 7.7 days (Czorlich 2023, PMID 37004138); after the primary neurosurgical diagnosis it ranged from 7 days to 5 years (Akins 2011, PMID 21523524).
- **Early warning:** falling EVD output at unchanged drain height as the state develops (Akins 2011); quantitative EEG slowing many hours before clinical deterioration, reversing with CSF removal (Wang 2024, PMID 39740216).

2. Pressure measurement

- **Direct ventricular measurement is required.** Lumbar pressure may not reflect ventricular pressure when the compartments are disconnected (Chang 2026 lumbar-ventricular discrepancy, PMID 42218510; Staykov 2011, PMID 21273925), and in post-LP cases the lumbar compartment is the sink. Shunt-tap manometry through the reservoir is acceptable in shunted patients (Hoffman 2026, PMID 42224725), as is Ommaya puncture (Chen 2025).
- **Reported values:** 2.2–6.6 mm Hg (Pang 1994); mean 4.8 mm Hg (Lesniak 2002); –3 mm Hg (Pandey 2017); –4 cm H₂O (Diaz-Romero Paz 2019); all < 70 mm H₂O and as low as –10 cm H₂O (Wu 2019); –5 to +10 mm Hg with excursions to –21 mm Hg (Salam 2025); ≤ 5 cm H₂O inclusion threshold in most series (Keough 2021; Datta & Tu 2025).

- **Continuous monitoring** (parenchymal or EVD transducer) is used to titrate therapy and to document the return of positive ICP as the end-point of sub-zero drainage (Pandey 2017; Godoy 2023 — drainage continued "with concomitant ICP monitoring at the neuro-ICU"). Background technique papers: Chari 2017 intraparenchymal monitoring for CSF disorders (PMID 28799016); Toma 2010 continuous ICP to investigate shunt function (PMID 20799857); Arroyo-Palacios 2016 ICP pulse morphology predicting EVD weaning (PMID 27106888); Eide 2012 showing that ICP pulse-wave parameters barely change when mean ICP is negative until it falls below -15 mm Hg (PMID 22153319) — so pulse amplitude cannot be used to detect the state.
- **The trap:** a normal ICP (5–12 mm Hg) in a patient whose shunt is set to 10 cm H₂O or whose EVD is at 15 cm H₂O is inappropriately low for the degree of ventriculomegaly and symptoms; the absolute number is less important than the mismatch between ventricles, symptoms and pressure (Hamilton & Price 2012; Gnanalingham 2004 "14 mmHg: a case of raised intracranial pressure?", PMID 15201374 — retained as a reminder that thresholds are individual).

3. Imaging

- **Ventriculomegaly** out of proportion to the sulci: Evans index > 0.3 (Duan & Hu 2024, PMID 38579413), rising frontal-horn ratio tracked as the treatment target (Houlden 2018, PMID 30009119), third-ventricular width used as the response metric (Bruk 2026, PMID 42732439: $13 \rightarrow 8$ mm and $13 \rightarrow 4$ mm).
- **Effaced convexity subarachnoid space and Sylvian cisterns** — the CSAS is collapsed because it is at lower pressure than the ventricles; the 2025 Chinese consensus makes this an explicit criterion, together with a rounded, enlarged fourth ventricle ("ball sign") implying fourth-ventricular outlet obstruction (PMID 40734409).
- **Transependymal / periventricular oedema despite low ICP** in 5/9 of Akins' patients — counter-intuitive and a pointer to the poroelastic model.
- **Enlarged perimedullary cisterns and brain-stem distortion** after LP-induced LPH (Vilela 2020, PMID 32712403), and descending transtentorial herniation with ventriculomegaly (Marianayagam 2017, PMID 28209310) — the caudal compartment is at lower pressure so structures are pulled down.
- **Absent cistern sign** and other LP-shunt function signs are unreliable (Kulkarni 1999; excluded).
- **Cine/phase-contrast MRI** documented CSF pathway obstruction in all 20 SILPAH patients (Hamilton & Price 2012) and reversed aqueductal flow toward the ventricles during LPH (Bateman 2026, PMID 42355921). Radionuclide or contrast ventriculography via the EVD has been used to localise the block (Rekate 2008; Filippidis 2011).
- **MR elastography** — very low shear stiffness during an episode, recovered two years later (Olivero 2016, PMID 27198914; Olivero 2021, PMID 33030603); Strand 2018 (PMID 28962963) suggests MRE as the tool most likely to give an objective diagnostic marker.
- **Sunken flap / paradoxical herniation** in craniectomised patients (Salam 2025; Bateman 2026).

4. Proposed diagnostic criteria (converging sets)

Criterion	Pang & Altschuler 1994	Duan & Hu 2024	Datta & Tu 2025	Chinese consensus 2025
Symptoms of hydrocephalus / neurological worsening	✓	✓	✓	✓ (progressive deficits)
Ventriculomegaly	✓	Evans > 0.3	change in ventricular size	progressive dilation; effaced cisterns; 4th-ventricle "ball sign"
Ventricular ICP	≤ 5 cm H ₂ O	≤ 5 cm H ₂ O	≤ 5 cm H ₂ O	0–70 mm H ₂ O on ventricular puncture
Response to drainage	improvement with sub-zero drainage	improvement with negative-pressure drainage	improvement after CSF drainage	(management criterion)

Criterion	Pang & Altschuler 1994	Duan & Hu 2024	Datta & Tu 2025	Chinese consensus 2025
Exclusions	patent shunt	ventriculomegaly from neurodegeneration	—	high-risk history required

Bakmeedeniya 2026 proposes a "diagnostic and management framework" but the field still lacks a validated definition.

5. Differential diagnosis

Entity	Shared features	Discriminators
Shunt obstruction / high-pressure hydrocephalus	identical symptoms, ventriculomegaly	ICP high on tap; shunt revision works. LPH: patent shunt at tap/revision, low ICP, revision fails (Filippidis; Hoffman)
Shunt overdrainage / slit-ventricle syndrome (Faulhauer 1978, PMID 742440; Serlo 1985/1989; Rekate 2008 shunt-related headaches, PMID 18259760; Pedersen 2023 definition review, PMID 36639536)	low or negative ICP, postural headache, subdural collections (McCullough 1974)	ventricles small or slit-like; Rekate type-1 SVS is the low-pressure form; treatment is raising resistance (anti-siphon, higher valve setting; Gruber 1984, PMID 6726390; Aschoff 1995, PMID 7621479) — the opposite of LPH. Bateman 2013 hypertensive SVS (PMID 23991845) shows SVS itself is heterogeneous
Spontaneous / iatrogenic intracranial hypotension (Mokri 2000, PMID 11113229; Binder 2002 intrathecal saline for obtundation, PMID 12188967; Samadani 2003 after lumbar drainage, PMID 12493112; Atkinson 1998 acquired Chiari, PMID 9452230)	postural headache, low ICP, sagging brain, obtundation, response to blood patch/Trendelenburg	ventricles small; no ventriculomegaly. LPH is best understood as intracranial hypotension plus a ventricle-SAS disconnection that inverts the gradient
Post-dural-puncture headache in a shunted patient (Koenig 2025, PMID 41357016)	post-LP, headache	check ventricular size — if enlarging, treat as LPH; blood patch carries extra considerations
Normal-pressure hydrocephalus (historical axis; Khan 2013 overdrainage in iNPH, PMID 23930853; Funnell 2020)	ventriculomegaly with normal opening pressure, gait/cognitive decline	chronic, elderly, B-waves and intermittent hypertension (Symon 1972); improves with standard shunting; "ultra-low-pressure NPH" is the overlap zone
Cerebral venous overdrainage (Barami 2016, PMID 27581321)	after CSF diversion, low ICP	may be a mechanism of LPH rather than a separate entity
Sinking-flap syndrome / paradoxical herniation	craniectomy, sub-atmospheric physiology, deterioration on LP or drainage	flap sunken, midline shift away from defect; cranioplasty is curative; LPH may coexist or follow (Bateman 2026; Bhatjiwale 2023)
External hydrocephalus of infancy (Sainz 2019, PMID 30474714)	infant macrocephaly	enlarged SAS with small ventricles from venous hypertension — the mirror image
Ventriculomegaly ex vacuo / neurodegeneration	large ventricles, normal ICP	no progression with symptoms; no response to drainage (Duan & Hu exclusion)
Tension pneumocephalus with skull-base leak (Stevens 2024, PMID 39220674)	post-shunt deterioration, leak	air on CT; both may coexist

6. Practical diagnostic sequence distilled from the algorithms (Keough 2021; Casado Pellejero 2022; Hoffman 2026; Duan & Hu 2024)

1. Suspect LPH in any patient with symptomatic ventriculomegaly and (a) a patent shunt, (b) a recent LP/lumbar drain/CSF leak/craniectomy, (c) an EVD that drains little at standard height, or (d) SAH/IVH with clinical stagnation at ~2 weeks.
2. Measure ventricular pressure directly (EVD, shunt reservoir, Ommaya).
3. Confirm shunt patency (tap, shuntogram) rather than revising empirically.

4. Image for the sink and the block: CT for ventricles, sulci, flap, pneumocephalus; MRI with cine/phase-contrast for aqueduct/fourth-ventricle outlet patency; look for pseudomeningocele, skull-base defect, spinal CSF collection.
5. Therapeutic test: lower EVD to 0 then sub-zero targeting 5–15 mL/h; clinical and radiological improvement confirms the diagnosis (Pang/Foster criterion 4). The series call the response "rapid" (Owler 2001) but give no fixed reassessment interval.
6. Exclude/treat coexisting infection, hyponatraemia and hypocapnia, which lower turgor.

Management — acute algorithm, definitive diversion, refractory disease

No randomised or prospective comparative data exist. What follows is the consensus pattern extracted from the series, algorithms and reviews (Pang 1994; Owler 2001; Akins 2011; Filippidis 2011; Hamilton & Price 2012; Hunn 2014; Foster 2016; Smalley 2017; Wu 2019; Keough 2021; Casado Pellejero 2022; Godoy Hurtado 2023; Duan & Hu 2024; Li 2024; Chen 2025; Datta & Tu 2025; Bakmeedeniya 2026; Chinese consensus 2025), with dissent noted.

1. Acute phase — restore brain volume

1a. Remove or counter the sink first where possible

- **Post-LP / lumbar drain:** flat or Trendelenburg positioning; stop the drain; epidural blood patch if not resolving in 24–48 h (Dias 1999, PMID 10352418; Smalley 2017, PMID 28362957; Vilela 2020 adult, PMID 32712403; Suzuki 2021 before EVD, PMID 35854689; Hoffman 2026, PMID 42224725; Koenig 2025 cautions on blood patch in shunted patients, PMID 41357016). In Smalley's 30 episodes post-LP cases needed a median 4 days of drainage vs 11 days for spontaneous cases.
- **CSF leak / pseudomeningocele:** direct repair, lumbar drain avoidance, fibrin sealant (Filippidis 2011, PMID 21800965; Stevens 2024, PMID 39220674; Kumar 2025 extra-axial ETV variant for persistent leak, PMID 41215506).
- **Cyst or spinal shunt:** ligate/remove (Vassilyadi 1995, PMID 7666227; Diaz-Romero Paz 2019, PMID 30710718; Ozeki 2022, PMID 34526456).
- **Sunken flap:** early cranioplasty (Bateman 2026, PMID 42355921; Zheng 2023 combined cranioplasty + shunt, PMID 36994490).

1b. Raise venous pressure / brain turgor (non-invasive, hours)

- **Neck wrapping / soft cervical collar / bilateral jugular compression:** Filippidis 2011 (all 3 patients), Pandey 2017 (PMID 28017745 — combined with Trendelenburg, hypoventilation to raise PaCO₂ and lowered plasma osmolarity to reverse ICP from –3 mm Hg), Bruk 2026 (PMID 42732439 — two children, third ventricle 13 → 8 and 13 → 4 mm within days, both then internalised), Kuppler 2025 (failed in chronic disease), Rekate 2019 review. Mechanism support: Hatt 2015 MRE (PMID 26045579). Risks: jugular thrombosis, discomfort, poor tolerance in agitated patients; requires ICP monitoring.
- **Head-down positioning / recumbency:** Foltz 1988; Dias 1999; Pandey 2017.
- **Fluid and gas management:** avoid hypocapnia and hyperosmolar therapy; Houlden 2018 (PMID 30009119) recommends avoiding fluid shifts; Pandey deliberately lowered osmolarity. This is expert opinion.

1c. Titrated external ventricular drainage (the reference standard)

Protocol synthesised from Pang 1994, Owler 2001 (PMID 11599454), Akins 2011 ("sub-zero method"), Hunn 2014 (PMID 24275338), Foster 2016 (PMID 27069743), Godoy Hurtado 2023 (PMID 37373809) and the paediatric reviews:

Each step is followed by the source wording it rests on (quotes from abstracts, or from the full text where marked FT — Foster, Godoy, Salam and Chen from the mirrored OA XML, Akins and Wu from publisher PDFs). Where the papers do not state a number, that is said explicitly rather than filled in. Akins 2011 is the only paper that lays out the sub-zero method as an explicit protocol; Wu 2019 gives no drainage settings at all; Pang 1994, Owler 2001 and Hunn 2014 are available here only as abstracts.

1. **Establish external ventricular access** — a new EVD or the existing shunt externalised; failure at conventional drainage pressure is part of the definition, not a step to be tried. - Pang 1994: "Seven patients failed to improve with multiple shunt revisions, including the use of low-pressure valves. In 11 patients, symptoms and ventriculomegaly were not reversed except with prolonged external ventricular drainage at subzero pressures". - Owler 2001: patients "failed to respond to apparently adequate CSF drainage via a functioning shunt (four cases) or external ventricular drain (one case)". - Foster 2016 (FT), Pang criteria: "(1) Neurologic decline at normal EVD or shunt pressures; (2) ventriculomegaly; (3) persistence of ventriculomegaly with ICP in the normal to low-normal range; and (4) clinical and radiographic response to sub-zero drainage". - Akins 2011 (FT) makes the failed conventional heights explicit in their inclusion criteria: "(4) clinical and radiographic response to EVD drainage at +5 to +10 cm height; (5) delayed neurologic decline accompanied by recurrent hydrocephalus despite confirmation of a patent EVD and normal ICP; (6) failure to respond to lowering EVD (0 to +5 cm); and (7) clinical and radiographic response to sub-zero drainage". Illustrated case 4: "EVD was lowered to +5 and then 0 cm without clinical or radiographic improvement." - Chen 2025 (FT), 28 paediatric cases with route recorded: "EVD, n (%) 12 (42.86%) ... VPS externalized, n (%) 16 (57.14%) ... Ommaya, n (%) 4 (14.29%)". - Salam 2025 (FT): "subatmospheric CSF diversion via external ventricular drain (EVD) placement or shunt externalization with or without neck wrapping". - Catheter site and calibre are not specified in any series; individual reports use a frontal EVD (Salam 2025: "a new right frontal EVD").
2. **Lower the drain below the reference point until CSF flows** — start at 0 to −5 cm H₂O, go to −10 if needed; deeper settings are reported only in refractory cases. No paper specifies a fixed interval between downward steps. - Akins 2011 (FT), describing Pang's method: "sub-atmospheric ventricular drainage achieved by setting the EVD height below the level of the external auditory canal"; their own practice: "sub-zero EVD drainage (typically, −3 to −10 cm)". Case 6 stabilised "with EVD at −3 cm height". - Owler 2001: "a period of external ventricular drainage at heights of 0 to −5 cm H₂O below the reference point (external auditory meatus—EAM)". - Foster 2016 (FT): "The EVD was initiated at a subatmospheric pressure, placed at a height ranging from 5 to 10 cmH₂O below midbrain"; the lowest setting in their illustrative case was "−12 cmH₂O". - Pang 1994: "mean external ventricular drainage nadir pressure of -5.7 ± 3.6 mm Hg" (≈ -8 cm H₂O). - Wu 2019 (FT): "Ventricular pressure was lower than 70 mm H₂O in all 39 patients, and the lowest value was −10 cm H₂O" — this is the lowest measured pressure ("Four patients showed the lowest pressure at −10 cm H₂O, 2 patients ... −5 cm H₂O, and 2 cases showed 0 cm H₂O in ventricle during EVD according to the 2 external auditory canal lines"), not a drainage setting. Wu's own management text says only "Precise CSF external drainage even at a subzero level" and cites Pang's sub-zero and Hunn's titrated methods as "innovative treatments"; no heights or steps are given. - Godoy Hurtado 2023 (FT): drainage "at negative pressures which were reached progressively to 15 cm below the EAC". - Salam 2025 (FT), refractory case: EVD "set to −8 cm H₂O", later "−10 cm H₂O", and at the extreme "set as low as −30 cm H₂O for forced ventricular reduction"; the drain "was adjusted hourly ... by either raising or lowering it by 2 cm H₂O" to allow spontaneous flow.
3. **Titrate to CSF output, not to a pressure** — 5–15 mL/h is the only explicit hourly target in a series; others give a daily volume. - Hunn 2014: "titrated to produce between 5 and 15 mL per hour of CSF drainage, including drainage under subatmospheric pressure if this was required to maintain CSF flow". - Salam 2025 (FT), summarising the literature: "large-volume aspiration of 30–50 mL of CSF and subsequent drainage of 10–15 mL/hr, with a willingness to lower drainage settings to less than zero to meet this target"; their own goals were 10, 15 and 15–20 mL/h. - Daily-volume authors: Godoy Hurtado 2023 (FT) "daily debts around 300 mL of CSF"; Pandey 2017 (PMID 28017745) "daily cerebrospinal fluid drainage >200 mL". - Akins 2011 describes the output signature rather than a target: "As patients entered the LPH state, the ICP remained in a normal range yet daily CSF output from the external ventricular drain was reduced."

4. **Expect a CSF "diuresis" when sub-zero drainage starts; monitor ICP if a parenchymal sensor is available.** The duration is given only as "several days"; Akins attribute the extra fluid to water leaving the parenchyma, not entering it. - Akins 2011: "When LPH patients were drained at sub-zero levels, daily CSF output exceeded baseline values for several days and then receded to baseline ... When LPH was treated with the sub-zero method, a 'diuresis' of CSF ensued." (FT): "During sub-zero treatment, we observed a rise in EVD output which we attribute to reduction in ventricular volume and reduction in brain extracellular fluid"; sub-zero treatment "will 'wring out' the sponge-like cortical mantle ... Water will flow from the parenchyma into the ventricular system." - Foster 2016 (FT): "We, too, noted decreased output during genesis of LPH with a significant increase in output during sub-zero drainage." - Godoy Hurtado 2023 (FT): "Continuous recording of ICP with the intraparenchymal sensor during subzero drainage showed negative pressures, a flat recording with no pathological waves, and minimal amplitude between systolic and diastolic values, suggesting a very distensible or 'compliant' brain."
5. **Response to sub-zero drainage confirms the diagnosis** (criterion 4 above). The series describe the response as rapid but do not fix a reassessment interval. - Owler 2001: "Four of the five patients showed rapid and significant clinical improvement ... In the fifth patient there was no improvement with low pressure external drainage and no shunt revision was undertaken." - Hunn 2014: "TEVD resulted in improvement in Glasgow Coma Scale (mean increase 4.6, $p=0.003$), and increases in ICP (mean increase 8.5)"; "Mean delay from first negative ICP to TEVD was 1.8 days." - Pang 1994: "During external ventricular drainage treatment, symptoms correlated only with ventricular size and not with ICP."
6. **Wean upward slowly once ventricles are small and CSF drains at a positive pressure; expect weeks to months.** Both explicit protocols use a 3–5-day step interval. - Akins 2011 (FT): "the EVD was raised in small increments (typically 3 cm steps) every 3–5 days. Prior to raising the EVD to a more positive height, clinical and radiographic stability was confirmed. If patients relapsed during EVD weaning, then the process was repeated. Shunt internalization was performed once clinical and radiographic stability was confirmed at neutral or positive EVD settings (typically 0 to +3 cm)." "Three patients had refractory LPH and were unable to tolerate positive pressure EVD drainage" and went to ventriculopleural or valveless shunts. - Foster 2016 (FT): the EVD was "incrementally raised 1–2 cmH₂O every 3–5 days"; sub-zero drainage continued "until the ventricular shrinkage was maintained at a positive pressure, and a new shunt system could be inserted"; "patients required prolonged hospitalizations for protracted sub-zero EVD, often on the order of months". Mean time from first ventriculostomy to shunt was 73 days (range 14–257) without ETV and 114 days (0–236) with ETV, $P = 0.16$ — means, not medians — and the authors attribute the lack of difference to "an inherent bias to perform ETV on patients who were refractory to sub-zero drainage". One patient "underwent rapid EVD wean (in 8 days) and did not require shunting". - Pang 1994: sub-zero drainage "for a mean period of 22.2 days". - Chen 2025 (FT), pooled paediatric data: "the average number of days of external CSF drainage was 41.89 days, with a minimum of 2 days and a maximum of 365 days" ($n = 28$). - Wu 2019 (FT): "the time interval from onset to last VP shunt achieved was 31–3880 days, with an average of 376 days" (29 survivors). Their shunt timing rule: "VP shunt should be considered as soon as the ventricular pressure rises above 50–60 mm Hg" (sic — almost certainly mm H₂O, given their 70 mm H₂O threshold and valve settings), and only when temperature, blood count and CSF in all ventricular compartments are normal and there is no pneumocephalus.
7. **Infection is both a precipitant and a complication of prolonged drainage, and predicts poor outcome.** Wu 2019 is the one series that states both directions explicitly (though not the phrase "vicious circle"); Salam 2025 only re-quotes Wu's numbers. - Wu 2019 (FT), infection as cause: "blood and infection can result in adhesion and block in the arachnoid space, which will hinder the pathway between the cortical arachnoid space and ventricle" → transmantle gradient → ventriculomegaly at low/negative pressure. Their diagnostic criterion 1 lists prior haemorrhage or posterior-fossa surgery "especially combined with CNS infection". - Wu 2019 (FT), infection as consequence: "external CSF drainage with a long time course, especially the negative pressure in ventricles prone to backflow from the atmosphere, can explain why there was such a high rate of CNS infection in our cases"; "it was very difficult to drain CSF externally under low/negative ventricular pressure, which was very easy to backflow and cause CSF infection". Case 2 developed pneumocephalus "during cerebrospinal fluid drainage backflow". Prevention: "Preventing backflow, strict aseptic manipulation, and reducing the drainage time were the key issues". Zero

pressure is classed with NegPH precisely because it carries "a high risk of backflow". - Wu 2019 (FT), burden: "Eighteen cases were complicated by intracranial infections ... 46.15%"; 87.5% (7/8) in NegPH vs 35.48% in LPH (P = 0.0149); of 8 deaths, "6 patients died from intracranial infections, which accounted for 75.0% and were the main reason for in-hospital death"; NegPH mortality 5/8 vs LPH 3/29 (P = 0.0056); mean GOS-E 1.63 (NegPH) vs 3.03 (LPH). - Akins 2011: "Conditions most commonly associated with LPH are: subarachnoid hemorrhage, chronic hydrocephalus, brain tumors, and chronic CNS infections." (FT): "We hypothesize that underlying inflammatory or degenerative changes alter periventricular brain permeability in LPH patients, for example, chronic CNS infection or chemical meningitis from subarachnoid hemorrhage." Conversely, they saw no infectious harm from the treatment itself: "No long-term complications were observed with low pressure shunting such as shunt infection or subdural hematoma." - Chen 2025 (FT), institutional cases drained via Ommaya for a median of 105 days: this approach "increase[s] the risk of intracranial infection, and one patient ultimately died from intracranial infection". - Bakmeedeniya 2026: "delayed intervention, infection, and irreversible white-matter injury predicted poor recovery". - Salam 2025 (FT): the index case developed refractory NegPH after shunt infection ("removal of his shunt" under ceftriaxone/ampicillin).

Dissent: Hamilton & Price 2012 and Casado Pellejero 2022 (PMID 34998487) prefer early ETV to avoid prolonged EVD; Chinese consensus and Li 2024 (PMID 38688300) recommend EVD/lumbar-cisternal drainage as first line with ETV or shunt for definitive care.

1d. Pharmacology (experimental)

- **Theophylline** (Salam 2025, PMID 40418891): refractory acute NegPH after TBI, hemispherectomy, shunt infection and IVH; ICP −5 to +10 mm Hg with excursions to −21 mm Hg despite sub-zero EVD (as low as −30 cm H₂O), two ETVs and ventriculopleural/ventriculoatrial shunts over ~4 months. After 48 h of oral caffeine without effect, "two doses of theophylline 200 mg twice a day" via PEG (borrowed from post-dural-puncture headache practice) were followed by spontaneous drainage at −18 cm H₂O and, on CT 4 h after the second dose, "complete collapse of the ventricles to an almost slit-like appearance"; focal seizures progressing to status epilepticus began 4 h after the first dose. Single case; mechanism (adenosine antagonism → cerebral vasoconstriction → restored CSF production / turgor?) untested.
- **Acetazolamide / diuretics** are contraindicated in principle (they lower ICP further); mentioned only as things to stop.
- No other drug has been reported for the entity; the consensus and review literature is silent on pharmacology beyond avoiding ICP-lowering agents.

2. Reconnecting the compartments — ETV

- **Rationale:** if the syndrome depends on ventricle-SAS disconnection, a fenestration abolishes the gradient and may abolish shunt dependence (Rekate 2008; Hamilton & Price 2012).
- **Data:**
- Hamilton & Price 2012 (PMID 22116443): 20 SILPAH patients, 10 managed with and 10 without ETV; ETV produced "a rapid return of normal ICP dynamics and a significant decrease in the number of shunt-dependent patients".
- Casado Pellejero 2022 (PMID 34998487): algorithm with early ETV; 5 patients, ETV alone succeeded in 50 %, remainder needed a shunt; one death from systemic infection.
- Villalonga 2025 (PMID 41165379): ETV in 16 adults, then valveless VP shunt if persistent; 81 % excellent outcome, mean KPS gain 56 points; 19 % system dysfunction requiring a valved shunt.
- Foster 2016 (PMID 27069743): 16 patients, ETV did not significantly reduce the duration of EVD or shunt dependence.
- Chen 2025 pooled paediatric: conservative measures and ETV rarely sufficed; 91 % ultimately required shunt repositioning/revision.
- Kuppler 2025: ETV failed in chronic NegPH.
- Kumar 2025 (PMID 41215506): "extra-axial ETV" (fenestration through the cranial defect) for LPH with persistent leak.

- **Interpretation:** ETV appears to help a subset — adults with demonstrable distal obstruction and preserved SAS absorption — and is a reasonable step once ventricles have been re-inflated; it is not a substitute for sub-zero drainage in the acute phase and its comparative benefit is confounded by selection (Godoy 2023).

3. Definitive CSF diversion — minimise resistance, exploit gravity

The permanent shunt must be able to operate at the sub-zero pressures the brain needs, i.e. the opposite design philosophy to anti-siphon devices:

Strategy	Sources	Notes
Programmable valve at lowest setting, no anti-siphon / gravitational unit	Pang 1994; Akins 2011; Marianayagam 2017; Chen 2025 (most children); Godoy 2023	Standard first choice; risk of later overdrainage/SVS so needs follow-up and gradual up-titration
Valveless VP shunt	Villalonga 2025; Badran 2018 (PMID 30287249 — improvised ultra-low-pressure valveless system); Duan & Hu 2024	Highest flow; watch for subdurals (McCullough 1974)
"Therapeutic siphoning" — non-siphon-control valve or adjustable gravitational valve set to permit upright siphoning	Bergsneider 1999 (PMID 10520938: adults improved with siphoning); Trinh 2013 (PMID 22770726: converting to siphon-permitting valve in shunted NPH with LPH physiology); Funnell 2020 (PMID 32989519: ultra-low-pressure state in NPH treated by siphoning)	Applies chiefly to adult NPH-overlap cases
Valve placed below the clavicle (caudal) to add a hydrostatic column	Carrera 2025 (PMID 39951714); Datta & Tu 2025 cite "shunt repositioning"	Converts body height into negative drainage pressure
Shunt into a negative-pressure cavity (pleura, atrium)	Salam 2025 (VPleural); Datta & Tu 2025; Chen 2025 (ventriculopleural in refractory children); Chiang 2001 (excluded — positive-pressure ventilation obstructs VPI shunts)	Pleural pressure −5 to −8 cm H ₂ O in spontaneous breathing; ineffective under mechanical ventilation
Wide-bore / low-resistance custom distal catheter	Galbarriatu 2013 (PMID 23904087 — thoracic-drain tubing as peritoneal catheter)	Reduces distal resistance
Active-pumping / negative-pressure shunt	Kalani 2013 (PMID 23380444 — investigational active shunt in a patient failing sub-zero EVD); background valve-technology literature: Aschoff 1995 (PMID 7621479), Czosnyka 2002 UK shunt evaluation (PMID 12111485), Chari 2014 (PMID 24405071)	Not commercially available; case-level evidence
Zero-ICP gravity shunt (ZIPS)	Foltz 1994 (PMID 8194062)	Historical; 56 patients, Foltz's own design
Intermittent reservoir pumping	Cheng 2017 (PMID 28264586: 7 adults with VP shunt at lowest setting; compressing the pump chamber several times daily improved ventricles in 6/7); Kuppler 2025 (PMID 40030163: scheduled active pressing as outpatient rescue)	Cheap, needs compliance and follow-up
Combined cranioplasty + shunt in craniectomised DOC patients	Zheng 2023	Single-stage restoration of container and drainage

Sequencing (Godoy 2023; Datta & Tu 2025; Chen 2025): internalise once the patient tolerates EVD raised to $\geq 0-5$ cm H₂O with stable ventricles; if internalisation fails at the lowest valve setting, consider valveless/negative cavity/caudal valve; if the patient re-develops LPH after a period of stability, re-externalise and repeat sub-zero drainage — many patients cycle several times (Wu 2019: 3.5 procedures per patient).

4. Chronic and refractory disease

Defined operationally as persistent or recurrent symptomatic ventriculomegaly after adequate sub-zero drainage, ETV and low-resistance shunting (Kuppler 2025's chronic negative-pressure hydrocephalus; Clarke 2006 / Houlden 2018 very-low-pressure hydrocephalus; Lin 2025 refractory hydrocephalus).

- **Predictors of refractoriness:** unsealed CSF leak, ongoing ventriculitis, suprasellar/skull-base surgery, multiple prior operations, severe underlying brain injury, permanently altered compliance (Houlden 2018; Bakmeedeniya 2026; Wu 2019).
- **Options tried:** prolonged (months) sub-atmospheric drainage; ETV; low-gradient shunt; neck wrapping (failed in Kuppler's case); intermittent active shunt pumping (Kuppler, Cheng); ventriculopleural conversion; theophylline (Salam); MRE-guided timing (Olivero — stiffness recovery may indicate when a standard valve can be tolerated); cranioplasty; and, in the DOC population, acceptance of a permanent low-pressure system with continued rehabilitation (Zheng 2023).
- **Prognosis:** Houlden's 4 VLPH cases and Wu's infected subgroup had the worst outcomes (mean GOS-E 2.7); Kuppler's patient remained shunt-dependent but stable as an outpatient.

5. Paediatric specifics

- Post-LP cases: recumbency, then blood patch; many resolve without drainage (Dias; Smalley).
- Spontaneous cases after posterior-fossa tumour surgery: EVD titration exactly as in adults; long durations (up to a year); most end with a programmable valve at lowest setting, some with ventriculopleural or repositioned shunts (Chen 2025).
- Neck wrapping is tolerated and effective in children (Bruk 2026).
- ETV alone is rarely sufficient (Chen 2025) — possibly because absorption at the convexity is also impaired by tumour/haemorrhage.
- Outcome: 77.5 % back to baseline; deaths driven by tumour progression.

6. What to avoid (from the failure reports)

- Standard-height EVD, standard valve settings or anti-siphon devices in a brain showing the paradox (McCullough 1986; Drake 1993; Trinh 2013).
- Empirical shunt revision without pressure measurement (Filippidis; Hoffman).
- Lumbar puncture or lumbar drainage in a patient with a patent shunt and obstructive pathology (Dias; Preuß; Ozeki) — or at least anticipate LPH.
- Osmotherapy, hyperventilation, diuretics in a patient with low ICP and large ventricles.
- Leaving a skull-base leak or pseudomeningocele untreated while draining sub-zero — the two sinks compete.
- Assuming that a "normal" ICP excludes hydrocephalus when the ventricles are enlarging and the patient is deteriorating.

Outcomes, prognostic factors, evidence quality and research agenda

1. Outcome data

Source	n	Outcome
Pang & Altschuler 1994 (PMID 7808607)	12	All improved with sub-zero drainage; most internalised with low-pressure hardware
Lesniak 2002 (PMID 12617236)	10	Ventricles returned to baseline size and stayed small at normal pressure after negative-pressure drainage

Source	n	Outcome
Akins 2011 (PMID 21523524)	9	Sub-zero method effective in all; internalised
Filippidis 2011 (PMID 21800965)	3	All resolved after leak repair + neck wrapping/drainage
Hamilton & Price 2012 (PMID 22116443)	20	Shunt dependence 20 % with ETV vs 70 % without
Hunn 2014 (PMID 24275338)	8	Mean GCS improvement 4.6; ventricular size reduced in all
Foster 2016 (PMID 27069743)	16	Prolonged EVD; ETV no significant benefit on EVD duration
Smalley 2017 (PMID 28362957)	29/30 episodes	Post-LP: median 4 days drainage, generally full recovery; spontaneous: median 11 days
Cheng 2017 (PMID 28264586)	7	Reservoir pumping improved 6/7
Houlden 2018 (PMID 30009119)	4 VLPH	Prolonged courses; recovery of GCS despite persistent ventriculomegaly in salvageable cases; poor overall
Wu 2019 (PMID 30611954)	39	Mean GOS-E 2.7 (1.6 NegPH vs 3.0 LPH); 8 deaths (5/8 NegPH vs 3/29 LPH), 6 of them from CNS infection
Keough 2021 SR (PMID 32736355)	195	Return to baseline 83 % children vs 49 % adults; overall mortality ~11 %
Casado Pellejero 2022 (PMID 34998487)	5	ETV alone 50 %; remainder shunted; low-pressure state resolved in all treated; 1 death (20 %) from systemic infection; 80 % good outcome
Czorlich 2023 (PMID 37004138)	15	Post-SAH aLPH → permanent shunt 66.7 % vs 17.4 % in other SAH patients; outcome otherwise not worse than matched SAH
Godoy Hurtado 2023 (PMID 37373809)	6	5/6 improved; treatment individualised; ~200 cases in literature
Chen 2025 pooled paediatric (PMID 40538927)	48	77.5 % returned to pre-existing status; deaths and poor outcomes due to underlying tumour/haemorrhage; mean external drainage 41.9 d
Villalonga 2025 (PMID 41165379)	16	81 % excellent, 19 % temporary improvement; mean KPS gain 56; 19 % needed a valved shunt
Datta & Tu 2025 (PMID 41191101)	6	All improved with drainage; shunt repositioning or negative-pressure cavities
Bakmeedeniya 2026 SR (PMID 41525875)	215	Prognosis governed by etiology and speed of recognition; standardised framework proposed

Summary: when recognised and treated with titrated sub-zero drainage, most patients regain their pre-episode status; mortality of ~10 % and poor functional outcome in about a quarter to a half of adults is attributable mainly to the underlying disease (SAH, TBI, tumour, ventriculitis) and to delayed recognition. Children and post-LP cases do best; infected and skull-base/VLPH cases do worst. Permanent shunt dependence is the norm (≥ 70 % without ETV, 50–80 % overall).

2. Prognostic factors reported

Favourable: paediatric age; post-LP precipitant; short delay to sub-zero drainage; identifiable and closable sink; retained viscoelasticity (GCS improves before ventricles shrink; Houlden); clear distal obstruction amenable to ETV.

Unfavourable: CNS infection; unsealed CSF leak; suprasellar/skull-base surgery; multiple prior operations; decompressive craniectomy with DOC; large IVH burden (Czorlich); need for drainage at ≤ -10 cm H₂O; chronic course > 3 months (Kuppler).

3. Complications specific to treatment

- Sub-zero EVD: ventriculitis with prolonged drainage; subdural haematoma or hygroma from overdrainage (McCullough & Fox 1974, PMID 4813717); intracranial hypotension symptoms; catheter blockage as ventricles collapse.
- Neck wrapping: jugular venous thrombosis risk, patient intolerance.
- Blood patch in shunted patients: theoretical risk of shunt tract/lumbar complications (Koenig 2025, PMID 41357016).
- Valveless / siphoning shunts: subdurals, slit ventricles — the pendulum swings to overdrainage (Faulhauer 1978, PMID 742440; Serlo 1985, PMID 4064018; Pedersen 2023, PMID 36639536); Doshi 2022 (PMID 36600762) illustrates oscillation between overdrainage and VLPH in one patient.
- Ventriculopleural shunts: pleural effusion; failure under positive-pressure ventilation.
- Theophylline: seizures (Salam 2025).

4. Evidence quality

- **Design:** 3 systematic reviews of case-level data, 1 consensus statement, ~24 retrospective series/cohorts (largest single-centre $n = 42$, Keough; largest pooled $n = 215$), 34 case reports, no prospective study, no trial, no registry.
- **Definitional heterogeneity:** ICP thresholds ≤ 0 , ≤ 5 cm H₂O, < 7 cm H₂O; some series include "normal" ICP with failed standard drainage; acute and chronic cases pooled.
- **Selection and publication bias:** successful sub-zero drainage and successful ETV are over-reported; unrecognised cases (likely frequent in DOC and SAH populations — Zheng 2023; Czorlich 2023) are absent.
- **Mechanistic evidence:** mostly inference from treatment response; one patient with serial MRE; one with phase-contrast CBF/CSF flow; no direct transmantle gradient measurements during an episode; animal models (Shulyakov 2012, Jugé 2016, Li 2008 kaolin) address hydrocephalus compliance generally, not LPH.
- **Outcome measures:** GCS, GOS-E, mRS, "return to baseline", frontal-horn ratio, third-ventricle width — inconsistent and rarely blinded.
- **Language/region:** a substantial body of Chinese literature (Wu 2019; Cheng 2017; Li 2024; Chen 2025; Zheng 2023; Chinese consensus 2025; Lin 2025) uses the < 70 mm H₂O threshold and reports higher prevalence in DOC/post-infectious cohorts than Western series — possibly true case mix, possibly definitional.

5. Open questions and a research agenda

1. **Definition:** adopt a single operational definition (symptomatic progressive ventriculomegaly + directly measured ventricular pressure ≤ 5 cm H₂O or ≤ 70 mm H₂O + failure of standard-pressure drainage), with acute/chronic and LPH/NegPH sub-labels — a Delphi process is the obvious next step after the 2025 Chinese consensus.
2. **Epidemiology:** prospective screening of SAH, posterior-fossa tumour, ventriculitis and craniectomy/DOC cohorts with routine ventricular pressure measurement when ventricles enlarge; the 3.7 % and 60 % figures need replication.
3. **Mechanism:** serial MR elastography and phase-contrast MRI during and after episodes; simultaneous ventricular and lumbar/cortical pressure recordings to test the transmantle hypothesis directly; interstitial-water quantification (diffusion, T2 mapping) to test the poroelastic model; venous sinus imaging to test the venous overdrainage/thrombosis association (Datta & Tu 2025).
4. **Treatment protocol:** standardise sub-zero EVD titration (height steps, output targets, weaning rate) and compare early ETV vs prolonged EVD in a multicentre registry; define when neck wrapping or blood patch can replace drainage.
5. **Hardware:** evaluate caudal valve placement, valveless systems, pleural shunts and active-pumping devices head to head for internalisation failure rates and overdrainage complications.

6. **Pharmacology:** methylxanthines (theophylline, caffeine) and other agents used for intracranial hypotension deserve a formal case series with ICP monitoring.
 7. **Biomarkers:** EVD output trajectory (Akins), quantitative EEG (Wang 2024), ICP pulse morphology (Arroyo-Palacios 2016; Eide 2012) as early-warning tools for the transition into the low-pressure state.
 8. **Long-term outcome:** cognitive and functional follow-up of survivors, incidence of later overdrainage/SVS after low-resistance shunting, and whether brain stiffness recovers (Olivero 2021) enough to permit conversion to standard valves.
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Key papers — per-paper extractions

Ordered

chronologically within groups. "FT" marks papers whose open-access XML is mirrored under `fulltext/`; the rest were read from Europe PMC abstracts.

A. Foundational descriptions

McCullough & Fox 1974 — Negative intracranial pressure hydrocephalus in adults with shunts and its relationship to the production of subdural hematoma (J Neurosurg; PMID 4813717). Shunted adults run sub-atmospheric ICP; subdural haematoma is the consequence. Establishes that negative ICP is achievable and harmful — the overdrainage pole of the same physiology.

McCullough 1986 — Symptomatic progressive ventriculomegaly in hydrocephalics with patent shunts and antisiphon devices (Neurosurgery; PMID 3491341). Patent shunts with ASDs; ventricles enlarge symptomatically; removing the ASD or lowering resistance helps. The paradox stated before it was named.

Foltz & Blanks 1988 — Symptomatic low intracranial pressure in shunted hydrocephalus (J Neurosurg; PMID 3343612). Upright, activity-related headache and lethargy; low upright ICP; relief when recumbent/Trendelenburg. Motivated the "zero-ICP" gravity-controlled shunt (Foltz 1994, PMID 8194062, 56 patients).

Pang & Altschuler 1994 — Low-pressure hydrocephalic state and viscoelastic alterations in the brain (Neurosurgery; PMID 7808607). 12 shunted patients, ventriculomegaly, ICP 2.2–6.6 mm Hg, patent shunts; prolonged sub-zero EVD restored ventricular size and symptoms; symptoms tracked ventricular size, not ICP. Proposes viscoelastic change; defines the entity and its treatment.

Vassilyadi, Farmer & Montes 1995 — Negative-pressure hydrocephalus (J Neurosurg; PMID 7666227). Child with a spinal arachnoid cyst-pleural shunt developing symptomatic ventriculomegaly at negative ICP; postulates a transmante gradient from the extra-ventricular sink. Coins "negative-pressure hydrocephalus".

Dias, Li, Freimark & Larson 1999 — Low-pressure shunt 'malfunction' following lumbar puncture in children with shunted obstructive hydrocephalus (Pediatr Neurosurg; PMID 10352418). Post-LP symptomatic ventriculomegaly with patent shunts; recumbency ± epidural blood patch resolved it. Establishes the paediatric post-LP phenotype and its benign treatment.

Bergsneider et al. 1999 — Beneficial effect of siphoning in treatment of adult hydrocephalus (Arch Neurol; PMID 10520938). Adults whose ventricles and symptoms improved only when the shunt was allowed to siphon — the "therapeutic siphoning" idea later applied to LPH.

B. Series that defined management

Owler et al. 2001 — Low pressure hydrocephalus: issues of diagnosis and treatment in five cases (Br J Neurosurg; PMID 11599454). Five cases; stresses direct ventricular pressure measurement and drainage at or below zero; first UK series.

Lesniak, Clatterbuck, Rigamonti & Williams 2002 — Low pressure hydrocephalus and ventriculomegaly: hysteresis, non-linear dynamics, and the benefits of CSF diversion (Br J Neurosurg; PMID 12617236). 10 adults, mean ICP 4.8 mm Hg; ventricles normalised only after negative-pressure drainage and then stayed small at normal pressure — a hysteresis loop.

Daniel, Lee & Halcrow 2002 — Low-pressure hydrocephalic state complicating hemispherectomy (Epilepsia; PMID 12027920). Post-hemispherectomy with CSF rhinorrhoea; ventriculomegaly at low ICP; treated with sub-zero drainage then low-pressure shunt.

Clarke & Meyer 2006 — Very low pressure hydrocephalus (J Neurosurg; PMID 16961147) and **Houlden et al. 2018** — Importance of frontal horn ratio and optimal CSF drainage in the treatment of very low pressure hydrocephalus (J Neurol Surg A; PMID 30009119). VLPH after suprasellar/skull-base resection is often refractory; Houlden's 4 cases suggest recovery of GCS despite persistent ventriculomegaly identifies salvageable patients ("retained viscoelasticity") and recommends frontal-horn-ratio tracking, volume optimisation and avoidance of fluid shifts.

Rekate, Nadkarni & Wallace 2008 — The importance of the cortical subarachnoid space in understanding hydrocephalus (J Neurosurg Pediatr; PMID 18590387). Reframes LPH as ventricle-CSAS disconnection with a lower-pressure CSAS and loss of brain turgor; predicts benefit from ETV, neck wrapping, and closing sinks. Companion: Rekate 2019 Low or negative pressure hydrocephalus demystified (PMID 31103762) and 2021 commentary (PMID 34272587).

Akins, Guppy, Axelrod, Chakrabarti, Silverthorn & Williams 2011 (FT, publisher PDF) — The genesis of low pressure hydrocephalus (Neurocrit Care; PMID 21523524). 9 neuro-ICU patients (0.2% of admissions; 4 of 102 SAH patients with EVDs); LPH 7 days–5 years after the index condition; EVD output fell as LPH developed; transependymal oedema in 5 despite low ICP. The only paper to state the sub-zero method as a protocol: response to EVD at +5 to +10 cm, then failure at 0 to +5 cm, then sub-zero drainage "typically –3 to –10 cm" below the external auditory canal; wean "in small increments (typically 3 cm steps) every 3–5 days" only after clinical and radiographic stability, repeat if relapse; internalise at 0 to +3 cm. Output rose above baseline for several days ("CSF diuresis"), attributed to water leaving the parenchyma (Darcy flux, poroelastic "boggy brain"). All 9 resolved; 6 programmable valves at lowest setting, 1 valveless VPS, 2 ventriculopleural (the 3 who could not tolerate positive EVD pressure); 2 relapses; no SDH or shunt infection at 6 months–4 years.

Filippidis, Kalani & Rekate 2011 — Negative-pressure and low-pressure hydrocephalus: the role of cerebrospinal fluid leaks resulting from surgical approaches to the cranial base (J Neurosurg; PMID 21800965). Three post skull-base-surgery cases with leaks and ventricle-SAS block; treated with leak repair, neck wrapping and drainage; formalises the leak-as-sink mechanism and neck wrapping as therapy.

Hamilton & Price 2012 — Syndrome of inappropriately low-pressure acute hydrocephalus (SILPAH) (Acta Neurochir Suppl; PMID 22116443). 20 patients (10 without, 10 with ETV); all had EVD with drainage below zero; cine MRI showed CSF pathway obstruction in all; ETV "reestablishes communication between the SAS and ventricles" with rapid return of normal ICP dynamics and significantly fewer shunt-dependent patients.

Kalani, Turner & Nakaji 2013 — Treatment of refractory low-pressure hydrocephalus with an active pumping negative-pressure shunt system (J Clin Neurosci; PMID 23380444). Investigational active shunt for a patient failing sub-zero EVD.

Galbarriatu et al. 2013 — Low-pressure hydrocephalus: a case report and review (Neurocirugia; PMID 23904087). Custom wide-bore peritoneal catheter (thoracic-drain tubing) to minimise distal resistance.

Trinh et al. 2013 — Shunt overdrainage or low-pressure hydrocephalus? (Neurosurgery; PMID 22770726). Shunted NPH patient with LPH physiology treated by converting to a siphon-permitting system — highlights the overdrainage/LPH diagnostic fork.

Hunn et al. 2014 — Successful treatment of negative pressure hydrocephalus using timely titrated external ventricular drainage and a low-pressure ventriculoperitoneal shunt (Clin Neurol Neurosurg; PMID 24275338). 8 adults, mean 3.0 prior procedures; TEVD titrated to 5–15 mL/h including sub-atmospheric drainage, then low-pressure VP shunt; mean GCS improvement 4.6; first series to report acceptable long-term outcomes.

Foster, Zlotnikov & Rekate (Foster 2016) — Analysis of low-pressure hydrocephalus: does ETV shorten time to internalization? (World Neurosurg; PMID 27069743). 16 patients; ETV did not significantly shorten EVD duration.

Olivero et al. 2016 / 2021 — Magnetic resonance elastography demonstrating low brain stiffness in a patient with low-pressure hydrocephalus (PMID 27198914) and Brain stiffness following recovery in a patient with an episode of low-pressure hydrocephalus (PMID 33030603). The only direct material-property data: very low shear stiffness during the episode, recovered two years later.

Barami, Sood 2016 — The cerebral venous system and the postural regulation of intracranial pressure: implications in the management of patients with cerebrospinal fluid diversion (Childs Nerv Syst; PMID 27581321). Venous overdrainage model for low-pressure states after CSF diversion.

Pandey, Ali, Vasenina & Prasad 2017 — Management of negative pressure hydrocephalus (Neurol India; PMID 28017745). ICP –3 mm Hg reversed with Trendelenburg, cervical wrapping, hypoventilation (raising PaCO₂) and lowered plasma osmolarity — the most explicit "raise ICP medically" protocol.

Smalley, Spiro & Dias 2017 — Low-pressure hydrocephalus in children: a case series and review of the literature (Neurosurgery; PMID 28362957). 30 episodes in 29 children; 37.9 % post-LP; post-LP cases: conservative 27 %, blood patch 18 %, drainage in the rest; median 4 vs 11 days of drainage for post-LP vs spontaneous.

Cheng et al. 2017 — Clinical manifestations and treatment of low-pressure hydrocephalus with ventriculoperitoneal shunt (Chinese; PMID 28264586). 7 adults with shunts at lowest setting still enlarging; intermittent manual compression of the reservoir several times a day improved 6/7 — the origin of the "reservoir pumping" strategy later used by Kuppler 2025.

Marianayagam et al. 2017 — Trapped fourth ventricle and LPH (PMID 28209310) and **Badran et al. 2018** — improvised ultra-low-pressure valveless shunt after SAH in haemangioblastomatosis (PMID 30287249): hardware improvisation case reports.

Strand, Balise, Leung & Boop 2018 — Low-pressure hydrocephalus: a case report and review of the literature (World Neurosurg; PMID 28962963). 25-year search, 17 articles, 90 cases; argues MRE may become the objective diagnostic tool and ETV a strong alternative/adjunct.

Wu, Zang, Wu, Sun, Yu & Hu 2019 (FT, publisher PDF) — Diagnosis and management for secondary low- or negative-pressure hydrocephalus and a new hydrocephalus classification based on ventricular pressure (World Neurosurg; PMID 30611954). 39 adults (27 TBI, 12 spontaneous haemorrhage) = 13% of 298 hydrocephalus cases at a Shanghai emergency centre, 2013–18; ventricular pressure < 70 mm H₂O in all, 8 at ≤ 0 (four at –10 cm H₂O); ~3.5 operations each; onset-to-final-shunt mean 376 d (31–3880). CNS infection 18/39 (46%), 87.5% in NegPH vs 35% in LPH (P = 0.015); 8 deaths, 6 from infection; NegPH mortality 5/8 vs 3/29; mean GOS-E 2.7 (1.6 NegPH vs 3.0 LPH). Attributes infection to prolonged drainage with atmospheric backflow at negative pressure, and infection/blood to arachnoid adhesion causing the transmantle gradient. Proposes diagnostic criteria (history, deterioration, CT "ball sign" of the fourth ventricle, effaced convexity/Sylvian/ambient cisterns, pressure < 70 mm H₂O) and a classification (high > 200, normal 70–200, low 0–70, negative ≤ 0 mm H₂O). Management is descriptive, not a protocol: "precise CSF external drainage even at a subzero level", neck wrapping/bellyband/Trendelenburg, nimodipine, VP shunt once pressure recovers and CSF is sterile in all compartments, no anti-siphon device, no LP shunt if craniocervical cistern effaced, and reservoir pumping hundreds of times a day for persistent ventriculomegaly.

Diaz-Romero Paz et al. 2019 — Negative-pressure hydrocephalus: a therapeutic challenge (PMID 30710718). ICP -4 cm H₂O after external drainage of a cyst; removal of the sink plus sub-zero EVD; transmante-gradient discussion.

Funnell et al. 2020 — The ultra-low-pressure hydrocephalic state in NPH: improvement after shunt siphoning (Acta Neurochir; PMID 32989519). Shunted iNPH patients who relapsed with large ventricles at very low ICP and improved when the valve was reset to allow siphoning; defines the NPH overlap.

Vilela et al. 2020 (FT) — Low-pressure hydrocephalus after lumbar puncture in an adult (PMID 32712403). Enlarged perimedullary cisterns and brain-stem distortion; epidural blood patch curative; venous overdrainage discussion.

Keough, Isaacs, Urbaneja, Hamilton et al. 2021 — Acute low-pressure hydrocephalus: a case series and systematic review of 195 patients (J Neurosurg; PMID 32736355). 42 local + 153 literature; 53 children, 142 adults; decreased LOC predominant; neoplasm and haemorrhage the commonest etiologies; 39 % of children post-LP; shunt in situ 92 % children / 39 % adults; sub-atmospheric drainage the main temporising step; shunt or ETV definitive; mortality 11 %; return to baseline 83 % children / 49 % adults.

Weisbrod et al. 2021 (FT) — Angiogram-negative SAH and low-pressure hydrocephalus after COVID-19 (PMID 34462697). Single case; adds a systemic-inflammatory trigger.

Casado Pellejero et al. 2022 — Acute negative-pressure hydrocephalus: management algorithm and value of early endoscopic third ventriculostomy (World Neurosurg; PMID 34998487). 5 patients; algorithm with early ETV applied in 4; ETV alone succeeded in 2 (50 %), 2 needed shunts; 1 death from systemic infection; 80 % good outcome.

Czorlich et al. 2023 — Acute low-pressure hydrocephalus in aneurysmal SAH (PMID 37004138). 15/406 (3.7 %); onset 13.1 ± 7.7 days; IVH presence/extent associated; aLPH patients went on to chronic hydrocephalus and shunting far more often than other SAH patients.

Godoy Hurtado et al. 2023 (FT) — Low- and negative-pressure hydrocephalus: new report of six cases and literature review (J Clin Med; PMID 37373809; PMC10299038). Six cases (medulloblastoma $\times 2$, TBI with bifrontal craniectomy, craniopharyngioma, leptomenigeal glioneuronal tumour, shunted NPH); ~ 200 cases in literature; symptoms follow ventricular size not pressure; sub-zero drainage most used; neck wrapping, ETV, blood patch as alternatives; pathophysiology "not clear" — permeability/viscoelasticity changes plus craniospinal SAS imbalance.

Zheng et al. 2023 — Hydrocephalus in severe disorders of consciousness (PMID 36994490). 68 DOC patients with ventriculomegaly; 60.3 % (41/68) had LPH/NegPH; 45.5 % had one-stage VP shunt + cranioplasty; individualised low valve pressure by ventricular size and consciousness scores.

Duan & Hu 2024 — Low-pressure hydrocephalus: a narrative review (PMID 38579413). Proposes criteria (Evans > 0.3 , ICP ≤ 5 cm H₂O, response to negative-pressure drainage, exclusion of neurodegeneration); risk factors include prior radiotherapy, infection, haemorrhage, multiple surgeries; treatment ladder EVD $\rightarrow$ ETV/valveless shunt.

Li et al. 2024 — Low/negative-pressure hydrocephalus: a review (PMID 38688300). Four mechanisms (transcortical pressure difference, venous overdrainage, viscoelastic change, porous elastic sponge); EVD/lumbar-cisternal drainage first, ETV or shunt definitive.

Wang et al. 2024 — Quantitative EEG as early warning in post-SAH acute low-pressure hydrocephalus (PMID 39740216). qEEG slowing preceded clinical deterioration and reversed with CSF removal.

Kuppler et al. 2025 (FT) — Chronic negative-pressure hydrocephalus after failure of prolonged drainage (PMID 40030163; PMC11877371). Refractory to low-gradient shunt, neck wrapping, ETV and prolonged sub-atmospheric EVD; scheduled intermittent active reservoir pressing enabled outpatient management; defines cNegPH.

Salam et al. 2025 (FT) — Theophylline for refractory acute negative-pressure hydrocephalus (PMID 40418891; PMC12105587). TBI, hemicraniectomy, shunt infection, IVH; ICP -5 to $+10$ mm Hg (nadir -21) despite ~ 4 months of sub-zero EVD (to -30 cm H₂O), two ETVs and ventriculopleural/ventriculoatrial shunts; two doses of theophylline 200 mg via PEG followed by spontaneous drainage, slit-like ventricles on CT at 4 h and sustained improvement; status epilepticus as complication. Single case.

Chen, Li et al. 2025 (FT) — Low- and negative-pressure hydrocephalus in children: a systematic analysis of 44 literature cases and 4 institutional cases (PMID 40538927; PMC12176887). Craniotomy (mostly posterior-fossa tumour) the leading antecedent; > 50 % prior surgery; fatigue, vomiting, headache, bradycardia, seizures, failure to thrive; mean external drainage 41.9 days (2–365); conservative measures (posture, reservoir pumping, valve adjustment) help a minority; 90.9 % need shunt repositioning; 77.5 % recover to baseline; poor outcomes track underlying disease.

Datta & Tu 2025 — Pediatric low-pressure hydrocephalus: a case series and re-analysis of a rare disease (PMID 41191101). 6 children, ICP ≤ 5 cm H₂O; low-resistance VP shunts (3) or negative-pressure cavities (3); prior jugular/sigmoid sinus thrombosis in 2/6; venous-thrombosis model.

Villalonga et al. 2025 — ETV plus valveless VP shunt hybrid for LPH (PMID 41165379). 16 adults meeting Pang–Altschuler criteria; mean KPS gain 56; 81 % excellent, 19 % temporary improvement; 19 % needed a valved shunt.

Carrera et al. 2025 — caudal (infraclavicular) programmable valve placement to exploit the hydrostatic column, 3 patients after prolonged sub-zero drainage (PMID 39951714).

Chinese Neuroscience Society / Neurosurgical Branch 2025 — Expert consensus on diagnosis and treatment of low-pressure hydrocephalus (PMID 40734409). Definition (< 70 mm H₂O, progressive ventriculomegaly, worsening deficits), imaging criteria (effaced Sylvian/convexity SAS, fourth-ventricle "ball sign"), candidate mechanisms (fourth-ventricular outlet obstruction, compliance change), management recommendations.

Bakmeedeniya et al. 2026 — Management of low/negative pressure hydrocephalus: a systematic review of etiology, pathophysiology and management (PMID 41525875). 30 studies, 215 patients (1980–2024); four interacting mechanistic models; management centred on sub-atmospheric EVD, low-pressure/valveless shunts, ETV; proposes a standardised diagnostic and management framework; prognosis governed by etiology and timely recognition.

Bateman & Bateman 2026 (FT) — Pathophysiology of sinking flap syndrome associated with low-pressure hydrocephalus (PMID 42355921). Phase-contrast MRI: bilaterally reduced CBF rising 26 % after cranioplasty; reversed aqueductal flow toward ventricles during LPH; lumped-parameter model favours autoregulatory reset over venous compression; hypothesises capillary CSF absorption at low ICP.

Bruk et al. 2026 (FT) — Neck wrapping for paediatric low-pressure hydrocephalus (PMID 42732439). Two children; third ventricle $13 \rightarrow 8$ mm and $13 \rightarrow 4$ mm; both internalised afterwards.

Hoffman et al. 2026 (FT) — Post-LP low-pressure hydrocephalus in a shunted adult (PMID 42224725). Diagnostic sequence: shunt-tap manometry, patency confirmation, imaging for sink and block; treated with drainage and blood patch.

C. Mechanistic and modelling anchors from the supporting axes

- Sklar 1980 brain elasticity changes with ventriculomegaly (PMID 6893601); Shapiro 1987 no transmante gradient in cats (PMID 3598675); Conner 1984 gradient in experimental NPH (PMID 6737056); Stephensen 2002 no gradient in communicating hydrocephalus (PMID 11904027); Eide & Saehle 2010 pulsatile gradient (PMID 20130957).
- Czosnyka 1996/2001 pressure-volume compensation (PMIDs 8778261, 11235954); Kim 2009 compensatory reserve (PMID 19240611); Gholampour 2022 compliance after shunting (PMID 35979170).

- Peña 1999, Dutta-Roy 2008, Chou 2016, Kim 2016 poroelastic FE models (PMIDs 10414573, 18534602, 26671218, 26274984); Solamen 2021 poroelastic MRE in NPH (PMID 32331966); Duy 2024 brain-CSF interface instability (PMID 38798141).
- Bateman 2002/2008 pulse-wave encephalopathy (PMIDs 12221445, 18379767); Egnor 2023/2026 windkessel impedance classification (PMIDs 37382303, 41791099); Nozaleda 2025 pulsation-driven glymphatic flow (PMID 40258946).
- Hatt 2015 jugular compression stiffens brain on MRE (PMID 26045579); Sainz 2019 venous hypertension and turgor (PMID 30474714); Shulyakov 2011/2012 and Jugé 2016 animal viscoelasticity (PMIDs 21316637, 22806081, 26848844); Wagshul 2021 reduced white-matter stiffness in early hydrocephalus (PMID 33631603); Dirrichs 2019 infant shear-wave elastography (PMID 31464808); Rekate 2020 infant compliance (PMID 32488353).
- Raimondi 1994 unifying theory (PMID 8194058); Oi 2011 multi-categorical classification (PMID 21928018); Orešković & Klarica 2011 bulk-flow critique (PMID 21641963); Bergsneider, Egnor et al. 2006 What we don't (but should) know about hydrocephalus (PMID 16572631).

08 • How the sub-zero EVD is actually titrated — source by source

Scope: what each paper says about (a) the starting height and how far/how fast it is lowered, (b) what is measured while drainage is sub-zero (ICP, output, CT, examination), (c) how "response" and "stability" are judged and how often, and (d) the rule for stopping the descent and starting the ascent. Quotes are verbatim; FT = full text read (OA XML mirrored in `fulltext/`, or the publisher PDF for Akins 2011 and Wu 2019); otherwise the abstract is the only source and that is said. Where a paper is silent on an item it is listed as "not stated" rather than inferred.

A. Sources that describe a titration method

Pang & Altschuler 1994 (PMID 7808607) — abstract only

- Setting / depth: "prolonged external ventricular drainage at subzero pressures (mean external ventricular drainage nadir pressure of -5.7 ± 3.6 mm Hg, for a mean period of 22.2 days)". The reference point (external auditory canal) and the "subzero method" name come from Akins' and Wu's descriptions of this paper, not from the abstract.
- What was measured: ICP (2.2–6.6 mm Hg at peak symptoms), ventricular/ biparietal ratio on CT, pressure–volume index in three patients.
- Response criterion: "During external ventricular drainage treatment, symptoms correlated only with ventricular size and not with ICP." Foster's restatement of Pang's method: "prolonged subatmospheric pressure drainage of the ventricular system until the ventricular shrinkage was maintained at a positive pressure, and a new shunt system could be inserted."
- Interval, step size, stop rule: not stated in the abstract.

Bergsneider 1999 (PMID 10520938) — abstract only

- "a protocol of progressive ventricular hypotension induced by external cerebrospinal fluid drainage"; "External ventricular drainage pressures as low as -30 cm H₂O were required to reduce ventricular size."
- Endpoint: symptoms "completely reversed once a normalization of ventricular size was achieved" — ventricular size, not a pressure, was the target.
- Step size and interval: not stated.

Owler 2001 (PMID 11599454) — abstract only

- "All five patients were tested for possible low pressure hydrocephalus by a period of external ventricular drainage at heights of 0 to -5 cm H₂O below the reference point (external auditory meatus—EAM)."

- Response: four of five improved clinically and radiologically; described as rapid. Interval, further lowering, stop rule: not stated.

Lesniak 2002 (PMID 12617236) — abstract only

- "All underwent intracranial pressure (ICP) monitoring and subatmospheric cerebrospinal fluid (CSF) drainage with an intraventricular or lumbar catheter." Mean ICP 4.8 mm Hg (0–10). "All patients improved only in the setting of negative pressure CSF drainage." No heights, steps or intervals.

Akins 2011 (PMID 21523524) — FT, publisher PDF

The only series that writes the method out as a protocol. - Entry criteria establish that positive and zero heights have failed first: "(4) clinical and radiographic response to EVD drainage at +5 to +10 cm height; (5) delayed neurologic decline accompanied by recurrent hydrocephalus despite confirmation of a patent EVD and normal ICP; (6) failure to respond to lowering EVD (0 to +5 cm); and (7) clinical and radiographic response to sub-zero drainage." - Depth: no starting height is stated. The only range given is the level at which patients had already responded — "Once LPH patients demonstrated clinical and radiographic improvement with sub-zero EVD drainage (typically, –3 to –10 cm), the EVD was raised in small increments" — so –3 to –10 is the effective range, not the opening setting. Reference point (from their account of Pang): "setting the EVD height below the level of the external auditory canal". Case 6 stabilised "with EVD at –3 cm height"; case 4 "was lowered to +5 and then 0 cm without clinical or radiographic improvement" before going sub-zero. - What was watched: daily EVD output (the paper's central observation — "As patients entered the LPH state, the ICP remained in a normal range yet daily CSF output from the external ventricular drain was reduced"; "When LPH patients were drained at sub-zero levels, daily CSF output exceeded baseline values for several days and then receded to baseline"), ICP via the EVD, and CT ("Neuroimaging was systematically reviewed"; transependymal fluid in 5, which "slowly resolved with sub-zero treatment"; "restoration of the subarachnoid space"). - Response / stability: "clinical and radiographic". Ascent rule: "the EVD was raised in small increments (typically 3 cm steps) every 3–5 days. Prior to raising the EVD to a more positive height, clinical and radiographic stability was confirmed. If patients relapsed during EVD weaning, then the process was repeated. Shunt internalization was performed once clinical and radiographic stability was confirmed at neutral or positive EVD settings (typically 0 to +3 cm)." - Stop-lowering rule: implicit — descend within –3 to –10 cm until output rises and the ventricles/exam improve; no explicit floor or interval between downward steps is stated. Three of nine "were unable to tolerate positive pressure EVD drainage" and went to ventriculopleural or valveless shunts.

Hunn 2014 (PMID 24275338) — abstract only

- Target is flow, not height: "titrated external ventricular drainage (TEVD), titrated to produce between 5 and 15 mL per hour of CSF drainage, including drainage under subatmospheric pressure if this was required to maintain CSF flow."
- Response measured: GCS ("mean increase 4.6, $p=0.003$ ") and ICP ("mean increase 8.5") — i.e. ICP rising back toward normal was itself treated as a marker of recovery. Ventricular size reduced in all eight.
- Reassessment interval, depth reached, stop rule: not stated.

Foster 2016 (PMID 27069743) — FT

- Start: "The EVD was initiated at a subatmospheric pressure, placed at a height ranging from 5 to 10 cmH₂O below midbrain" (note: midbrain, not EAM, as the reference). Lowest setting in the illustrated case "–12 cmH₂O".
- Ascent and its gate: "incrementally raised 1–2 cmH₂O every 3–5 days. Prior to elevating the drain, a computed tomography (CT) scan was obtained to document stable or smaller ventricular caliber. If the patient demonstrated neurologic decline, the drain was again lowered to a more negative pressure and the cycle reinitiated."

- Endpoint: "Patients underwent shunt placement once they demonstrated clinical and radiographic stability with an EVD leveled at or above midbrain." Illustrated case shunted "with the EVD at a pressure of +5 cmH₂O".
- Output: "We, too, noted decreased output during genesis of LPH with a significant increase in output during sub-zero drainage." No numeric target.
- Descent interval and stop-lowering rule: not stated beyond the 5-10 cm starting range; the paper is about ETV, not the titration.

Pandey 2017 (PMID 28017745) — abstract only

- Case treated "under continuous intracranial pressure monitoring" from −3 to +5 mm Hg with "daily cerebrospinal fluid drainage >200 mL" plus positioning (head down 10-15°), neck/chest/abdomen binding, hypotonic fluids (Na 130-135) and PaCO₂ > 40. Height, steps, CT interval: not stated.

Diaz-Romero Paz 2019 (PMID 30710718) — abstract only

- "the EVD level was adjusted to produce between 5 and 10 mL/hour of cerebrospinal fluid under a subatmospheric pressure set at −5 cm H₂O, and gradually raised in increments of 1 cm every 3 days until a positive ICP occurred." Endpoint = return of positive ICP; response = "excellent clinical and radiographic outcomes". CT interval: not stated.

Wu 2019 (PMID 30611954) — FT, publisher PDF

- No titration protocol. Management text: "Precise CSF external drainage even at a subzero level, maintaining proper systemic effective blood volume, and expanding cerebral blood vessels with nimodipine ... Neck wrapping, bellyband, horizontal head position, and even the Trendelenburg position can be used to improve ventricular pressure." They cite "the subzero method ... the titrated EVD method, and the active pumping negative-pressure shunt system" as others' innovations and conclude "there are still no standardized treatments".
- What they measured (diagnostic, not titration): "ventricular pressure was determined by the average height of the cerebrospinal fluid (CSF) water column when patients were in a quiet supine position and the head was flat for at least 15 minutes" — lowest −10 cm H₂O (four patients), −5 (two), 0 (two), "according to the 2 external auditory canal lines".
- Stop / shunt rule: "VP shunt should be considered as soon as the ventricular pressure rises above 50-60 mm Hg" (sic; almost certainly mm H₂O) and only with normal temperature and blood count, sterile CSF "in all ventricles and the arachnoid space", and no pneumocephalus.

Godoy Hurtado 2023 (PMID 37373809) — FT

- Depth: "four were treated with EVD, draining at a height of between 5 and 15 cm below the external auditory canal (EAC) for several days"; one case "responded to EVD draining at negative pressures which were reached progressively to 15 cm below de EAC, with daily debts around 300 mL of CSF" (the abstract's "zero to −15 mmHg" is the same range mislabelled in mm Hg).
- Measured: "Intermittent ICP measurements with closed drains ranged from −2 mmHg to zero"; in two patients an intraparenchymal sensor ("Continuous recording of ICP with the intraparenchymal sensor during subzero drainage showed negative pressures, a flat recording with no pathological waves, and minimal amplitude between systolic and diastolic values"); serial imaging.
- Endpoint / ascent: "It took 25 days at intensive unit care to recover normal ventricular size associated with level of consciousness maintained. Then level of EVD was progressively raised until Monro level. On day 40 ... new shunt". Relapse seven days after shunting → second catheter and "start again negative drainage" for 20 days. Overall: drain "until ventricular size normalized". Step size and CT interval: not stated.

Kuppler 2025 (PMID 40030163) — FT

- Failure at ≥ 0 first: "it was not possible to drain a sufficient amount of CSF when the EVD was positioned at an atmospheric pressure level ≥ 0 cm H₂O".

- Start and target: "subatmospheric CSF drainage was performed via the EVD, starting at a level of –10 cm below the external auditory canal (EAC) with a drainage rate of 5–15 ml/hr."
- How response was judged: "The efficacy of drainage was determined clinically by monitoring the patient's consciousness, speech production, and vegetative symptoms such as nausea and vomiting. In addition, it was ensured that an adequate amount of CSF was drained per day, and serial cranial CT scans were obtained to assess changes in ventricle size." "an average drainage of 160 ml CSF per day was needed to maintain a stable ventricle size."
- Planned ascent: "raising the draining chamber by 2 cm at 3-day intervals when the clinical parameters were stable" — which failed over 180 days (chronic NegPH); EVDs exchanged every 4 weeks for infection prophylaxis, and the patient still developed *S. epidermidis* ventriculitis.

Salam 2025 (PMID 40418891) — FT (refractory single case)

- Sequence of settings (index LPH episode; the EVD's height before –8 is not stated, only "CSF drainage was limited with ICPs less than 5 mm Hg"): "The EVD was set to –8 cm H₂O with a drainage goal of 10 mL/hr" → "forced hourly drainage for a minimum goal of 10–20 mL/hr" → new EVD "set at –10 cm H₂O with a goal drainage of 15 mL/hr" → "–5 cm H₂O, with a minimum drainage goal of 10 mL/hr" → "His EVD was set as low as –30 cm H₂O for forced ventricular reduction with a maximum ICP of –10 mm Hg. The ventricular size eventually reduced compared to weeks prior but still was larger than his prehospital baseline" (CT legend: "improvement in hydrocephalus ... with EVD adjusted to –25 cm H₂O") → "gradually raised to –10 cm H₂O, which was maintained for the following week" → later "weaned to 0 cm H₂O" → "a drainage rate of 15–20 mL/hr was targeted by lowering the drain to whatever negative pressure achieved that goal ... The drain's height was adjusted hourly to allow spontaneous CSF flow by either raising or lowering it by 2 cm H₂O."
- Measured: ICP continuously via EVD (–21 to +10 mm Hg over the course), hourly output, CT after each change of strategy (figures 2–5 are all CTs).
- Their literature summary of the method: "initially serial 1-hour ICP checks are recommended following shunt externalization or placement of an EVD. Once low or negative ICP is confirmed, aggressive measures must be undertaken to reduce the ventricular size, including large-volume aspiration of 30–50 mL of CSF and subsequent drainage of 10–15 mL/hr, with a willingness to lower drainage settings to less than zero to meet this target."

Lin 2025 (PMID 40826122) — FT, narrative review (Chinese practice)

- "Significant variation exists in the initial external ventricular drain height setting, ranging from –20 to –5 cmH₂O, achieved by one or more large-volume ventricular cerebrospinal fluid aspirations (30–50 ml), followed by drainage of cerebrospinal fluid at a fixed volume of 10–15 ml per hour." This is a secondary statement: the in-text citation number does not resolve to an LPH paper in the reference list, and no primary source in this corpus reports a starting height of –20 (the deepest documented starting settings are –10, Kuppler; –30 appears only as a floor reached later in Bergsneider and Salam). The 30–50 mL aspiration + 10–15 mL/h wording matches Salam's literature summary.
- Baseline imaging after response, then ascend: "After the patient is clinically stable and the ventricles have shrunk in size, a CT or MRI (e.g., rapid sequence MRI) scan of the head is performed to document the baseline ventricular size. The cerebral compliance is adjusted by gradually increasing the drainage strength (high) (1–2 cm every 3–5 days), and a stepwise decision is made when the EVD drainage height reaches above the level of the external auditory canal (≥ 0 cmH₂O, with some authors suggesting 4 cmH₂O), and clinical improvement is stabilized."

B. Single cases / smaller reports with a usable detail

- Doshi 2022 (PMID 36600762, FT): "lowering the EVD bag sequentially (in 5 cm gradient) below the reference level of tragus, until, there was a clinical and radiological improvement. Moreover, then, again gradually, raising it up (5.0 cm/day), till the negative or low-pressure state is reversed

back. Once the ICP values reached up to the 9.0 mmH₂O, with documented improvement in patients clinioradiological state" → shunt revised. (The 5 cm/day ascent is the fastest reported; a single case.)

- Galbarriatu 2013 (PMID 23904087, abstract): "maximum negative pressure of −8 cm H₂O".
- Weisbrod 2021 (PMID 34462697, FT, post-SAH): "Only once her EVD was progressively lowered to 0 cm H₂O did her mental status improve and ventricular size begin to decrease on follow-up CT imaging"; weaned and removed by post-bleed day 15 without a shunt — a response at zero, not sub-zero.
- Wang 2024 (PMID 39740216, FT, post-SAH): levels in mm Hg (15 → 0 → 10 → 0 → 5); each rise produced falling output (172 → 51 mL/day), rising delta power and a falling alpha/delta ratio on quantitative EEG before the examination changed, and each fall in level reversed it — the only report using qEEG as a continuous response marker. Manual drainage of ~30 mL when the level was dropped.
- Bruk 2026 (PMID 42732439, FT): "while draining, the patient was alert with Glasgow Coma Scale (GCS) 15/15, whereas clamping produced drowsiness (GCS 13/15) and recurrent ventriculomegaly"; ICP ~4 mm Hg throughout. States the general problem: "its protocol is not standardized, and drainage timelines are highly variable ... ventricular re-enlargement and neurological decline on each attempt to raise the drain".
- Chen 2025 (PMID 40538927, FT, paediatric pooled): no heights; drainage 41.89 days (2–365); institutional cases via Ommaya puncture, median 105 days, with CT before/after each stage; one death from infection attributed to the long drainage.
- Casado Pellejero 2022, Hamilton & Price 2012, Keough 2021 (abstracts): algorithms exist but the abstracts give no heights, steps or intervals; Hamilton/Price and Casado favour early ETV to shorten sub-zero drainage.

C. When was the drain first raised? Gating decisions per source

The question is what had to be true before the first upward step, and what sent the drain back down. Quotes are verbatim; FT = full text read, abs = abstract only.

Akins 2011 — FT (9 pts)

- Gate to begin ascent = response at sub-zero: "After clinical and radiographic improvement with sub-zero EVD drainage (typically, −3 to −10 cm), the EVD was raised in small increments (typically 3 cm steps) every 3–5 days."
- Gate for each further step: "Prior to raising the EVD to a more positive height, clinical and radiographic stability was confirmed."
- Reverse rule: "If patients relapsed during EVD weaning, then the process was repeated."
- End-point: "Shunt internalization was performed once clinical and radiographic stability was confirmed at neutral or positive EVD settings (typically 0 to +3 cm)."
- Case 6 shows the ascent gate being skipped for a refractory patient: "After sustained improvement with EVD at −3 cm, a valveless ventriculoperitoneal shunt was placed" — i.e. stability at a still-negative height was accepted as the end-point. "Three patients had refractory LPH and were unable to tolerate positive pressure EVD drainage" → ventriculopleural or valveless VP shunt.
- What "radiographic" meant: figures show CT at each phase (ventricular size, transependymal fluid, "restoration of the subarachnoid space"); no interval stated, no volumetric threshold. Clinical = exam/consciousness; autonomic signs and tachyarrhythmias are listed as presenting features that "resolved following LPH diagnosis and sub-zero treatment".

Foster 2016 — FT (16 pts)

- Gate for every upward step is a CT: "incrementally raised 1–2 cmH₂O every 3–5 days. Prior to elevating the drain, a computed tomography (CT) scan was obtained to document stable or smaller ventricular caliber."
- Reverse rule is clinical: "If the patient demonstrated neurologic decline, the drain was again lowered to a more negative pressure and the cycle reinitiated."

- End-point: "Patients underwent shunt placement once they demonstrated clinical and radiographic stability with an EVD leveled at or above midbrain." Illustrated case: CT "with external ventricular drain at 5 cmH₂O above midbrain" 3 h before the shunt.
- The trigger for ETV was failure of the ascent, not a scheduled step: "an inherent bias to perform ETV on patients who were refractory to sub-zero drainage and, importantly, all but 1 patient undergoing ETV was at sub-zero (one at zero) pressures at the time of ETV. Conversely, patients who demonstrated response to drainage were not readily considered for ETV." Time from LPH to ETV "ranged from 28 days to 6.5 months".
- Frame of reference for the whole cycle: sub-zero drainage "until the ventricular shrinkage was maintained at a positive pressure, and a new shunt system could be inserted."

Godoy Hurtado 2023 — FT (6 pts, 4 with EVD)

- Gate = two conditions together, ventricles and exam: "In cases of EVD, once normal ventricular size and a good clinical condition had been achieved, the drainage height was progressively increased by 2 cm/day until positive pressures were reached."
- Gate for the shunt: "If the clinical condition was maintained without deterioration, surgery was programmed for the placement of a definitive shunt".
- Time to reach the gate in the illustrated case: "It took 25 days at intensive unit care to recover normal ventricular size associated with level of consciousness maintained. Then level of EVD was progressively raised until Monro level. On day 40 after the onset, he could receive a new shunt". Relapse 7 days after shunting → "start again negative drainage. For 20 days he was maintained with the double catheter system."
- The ceiling can be found by failure: "Once basal size was recovered, the patient was unable to tolerate pressures above 2 cm of water, above which she would fall back into an arreactive coma. The whole time with forced negative pressure was 15 days." Another patient "needed to remain at zero to stay awake and return to baseline ventricular size."
- Monitoring during the cycle: "concomitant intracranial pressure monitoring at the neurointensive care unit" (EVD or intraparenchymal sensor) and serial imaging; no interval stated.

Kuppler 2025 — FT (1 case, chronic NegPH, ascent failed)

- Gate = clinical stability, checked against a fixed calendar: "The original treatment strategy was to achieve EVD weaning by raising the draining chamber by 2 cm at 3-day intervals when the clinical parameters were stable."
- What "clinical parameters" were: "The efficacy of drainage was determined clinically by monitoring the patient's consciousness, speech production, and vegetative symptoms such as nausea and vomiting. In addition, it was ensured that an adequate amount of CSF was drained per day, and serial cranial CT scans were obtained to assess changes in ventricle size."
- Reverse rule requires clinical and CT concordance: "In the event of clinical deterioration, correlating with radiographically confirmed ventricular CSF accumulation, the drainage chamber was lowered again."
- When the 3-day cadence kept failing the interval, not the step, was changed: "After an already prolonged clinical stay, the interval in which the draining chamber was raised was extended to 7 days. However, this approach did not lead to successful weaning from the EVD to a pressure level that would have allowed placement of a definitive low-pressure VPS" — final "necessary drainage level of -6 cm below the EAC" after 180 days, managed by intermittent reservoir pumping instead.

Diaz-Romero Paz 2019 — abs (1 case)

- Gate for the ascent is not stated beyond flow at -5; the ascent itself is scheduled and ends on ICP: "set at -5 cm H₂O, and gradually raised in increments of 1 cm every 3 days until a positive ICP occurred. Once clinical and radiographic stability was accomplished, a programmable ventriculoperitoneal shunt was inserted set to 30 mm H₂O."

Hunn 2014 — abs (8 pts)

- No ascent rule in the abstract; recovery is expressed as ICP rising with the exam: "TEVD resulted in improvement in Glasgow Coma Scale (mean increase 4.6, $p=0.003$), and increases in ICP (mean increase 8.5 ...)" followed by "subsequent low-pressure ventriculoperitoneal shunting."

Salam 2025 — FT (1 refractory case)

- Every rise in this case followed a CT showing smaller ventricles, and every rise was reversed by a CT showing re-enlargement or by loss of spontaneous flow:
- after -30: "The ventricular size eventually reduced compared to weeks prior but still was larger than his prehospital baseline (Fig. 3C). The EVD was gradually raised to -10 cm H₂O, which was maintained for the following week."
- after VA shunt + ETV: "His EVD was weaned to 0 cm H₂O with forced drainage oscillating between a minimum of 10 and 15 mL/hr to meet the drainage goal. Ventricular caliber did not improve, and over the next 2 weeks, a drainage rate of 15-20 mL/hr was targeted by lowering the drain to whatever negative pressure achieved that goal".
- the operative gate the authors actually used was spontaneous flow at a given height: "The drain's height was adjusted hourly to allow spontaneous CSF flow by either raising or lowering it by 2 cm H₂O."
- after theophylline: "CT demonstrated sustained hydrocephalus improvement, and his EVD drainage goal was decreased to 7 mL/hr, the lowest since admission. He was extubated 1 day later with his EVD at -5 cm H₂O and no drainage. The EVD was removed the following day (PHD 111)". Note the removal gate here was no drainage at -5 with a stable CT, not a positive height.
- Their literature summary of the end-point: "Once stabilized, definitive treatment must be established around one or a combination of the following: weaning the EVD, ETV, and/or ventricular shunt placement."

Lin 2025 — FT, narrative review

- Gate = clinical stability plus shrunken ventricles, then a documented baseline scan before the first rise: "After the patient is clinically stable and the ventricles have shrunk in size, a CT or MRI (e.g., rapid sequence MRI) scan of the head is performed to document the baseline ventricular size. The cerebral compliance is adjusted by gradually increasing the drainage strength (high) (1-2 cm every 3-5 days), and a stepwise decision is made when the EVD drainage height reaches above the level of the external auditory canal (≥ 0 cmH₂O, with some authors suggesting 4 cmH₂O), and clinical improvement is stabilized."

Doshi 2022 — FT (1 case)

- Gate = "clinical and radiological improvement", then a fast ascent ended on a measured ICP: "lowering the EVD bag sequentially (in 5 cm gradient) ... until, there was a clinical and radiological improvement. Moreover, then, again gradually, raising it up (5.0 cm/day), till the negative or low-pressure state is reversed back. Once the ICP values reached up to the 9.0 mmH₂O, with documented improvement in patients clinioradiological state" → shunt revision.

Other reports

- Pang 1994 (abs): no ascent rule; the abstract's only relevant observation is that "symptoms correlated only with ventricular size and not with ICP" during EVD — i.e. the thing to watch is the CT, not the pressure.
- Owler 2001 (abs): response ("rapid and significant clinical improvement") led directly to "shunt revision (three) or insertion (one)"; no ascent described.
- Bergsneider 1999 (abs): "completely reversed once a normalization of ventricular size was achieved ... All patients finally received a shunt"; no ascent described.

- Wu 2019 (FT): no EVD ascent rule at all. Their gate is for the shunt: "as soon as the ventricular pressure rises above 50–60 mm Hg [sic], at which a programmable VP will work well", plus normal temperature, normal blood leukocytes/neutrophils, "CSF tests in all ventricles and the arachnoid space should be normal" (including isolated chambers), and "pneumocephalus should not be present."
- Weisbrod 2021 (FT, post-SAH, responded at 0): "Her EVD was ultimately successfully weaned and removed on post-bleed day 15" — no gate stated.
- Wang 2024 (FT, post-SAH): the drain was raised for a complication, not for recovery — hygromas "attributed to excessive CSF drainage; therefore, the EVD level was raised on PBD 4 to 10 mm Hg" — and each rise was followed by rising qEEG delta power "for another 24 hours" before the exam changed, then re-lowering.
- Bruk 2026 (FT): the general problem — "Patients with LPH frequently cannot be weaned to physiologic pressures, with ventricular re-enlargement and neurological decline on each attempt to raise the drain". Their withdrawal gate for the neck collar (not the EVD) is the most explicit two-condition rule in the corpus: "discontinued only when two conditions were met together: neurological stability sustained for at least one week, and CT demonstrating improvement in ventricular caliber that was itself confirmed stable on subsequent imaging rather than on a single favorable scan." EVD removal in case 2: "An EVD was clamped for three days without any requirement for unclamping; measured ICP remained under 10 mmHg throughout, and level of consciousness was preserved, after which the drain was disconnected and removed."
- Hamilton & Price 2012, Keough 2021, Casado Pellejero 2022 (abs): no gating detail beyond favouring early ETV.

Comparison: what gates the first rise

Gate	Sources requiring it before the first upward step
Clinical improvement (consciousness/GCS ± speech, vegetative signs)	Akins, Foster (decline = reverse), Godoy ("good clinical condition"), Kuppler ("clinical parameters stable"), Lin ("clinically stable"), Doshi, Salam (exam + flow)
CT showing smaller/normal or stable ventricles	Akins ("radiographic improvement/stability"), Foster (CT before every step), Godoy ("normal ventricular size"), Kuppler (serial CT), Lin (documented baseline scan), Doshi, Salam (each rise after a CT)
ICP / pressure value	None as a gate to start rising; ICP is an end-point only (Diaz-Romero "until a positive ICP", Doshi 9 mm H ₂ O, Hunn +8.5 mean, Wu > 50–60 for shunt)
CSF output	Kuppler ("adequate amount ... per day"); Salam (spontaneous flow at the new height, later "no drainage at –5" before removal)
Fixed time at nadir	None; the cadence (3–5 d, Kuppler 3 → 7 d) starts only once the gate is met
Reverse trigger	Clinical decline (Foster "neurologic decline"; Akins "relapsed"); clinical decline confirmed by CT re-enlargement (Kuppler); "fall back into an arreactive coma" above +2 (Godoy). Salam (single refractory case) additionally lowered when CT showed "ventricular caliber did not improve" during a wean to 0; the hourly ±2 cm for spontaneous flow is a micro-adjustment around the set height, not a reversal of the ascent.
How far back down	Not specified by any source. Foster: "lowered to a more negative pressure and the cycle reinitiated"; Akins: "the process was repeated"; Kuppler: "the drainage chamber was lowered again"; Salam: "to whatever negative pressure achieved that goal". None says the drain was returned to the original nadir or to the last tolerated height; only the stepwise ascent (with its clinical + CT gate) is explicitly restarted.
End-point that replaces the ascent in refractory cases	Stability at a still-negative height → valveless VP / ventriculopleural shunt (Akins, –3 cm); ETV when the patient stays sub-zero for weeks to months (Foster, Kuppler); reservoir pumping at –6 (Kuppler); shunt with antigravity unit at zero (Godoy)

Bottom line on the ascent decision

1. Two things had to be true in every source that states a rule: the patient had improved clinically and a CT showed the ventricles smaller or normal (Akins, Foster, Godoy, Kuppler, Lin). Nobody starts rising on time alone, on output alone, or on a pressure value.

2. The CT requirement is explicit and per-step in Foster ("prior to elevating the drain, a CT scan was obtained"), implicit per-step in Akins ("radiographic stability was confirmed"), a one-off baseline in Lin, and "serial" in Kuppler and Godoy. Bruk's collar rule (one week stable plus a CT improvement confirmed on a second scan) is the strictest formulation but is not an EVD rule.
3. The clinical variable is level of consciousness in all; Kuppler adds speech and nausea/vomiting; Wang shows qEEG delta power lagging the exam by ~24 h after each rise.
4. Pressure enters only at the end: the ascent stops and a shunt is placed when the exam and CT stay stable at 0 to +3 cm (Akins), at/above midbrain (Foster), at "positive pressures"/Monro level (Godoy), ≥ 0 or +4 (Lin), or when a positive ICP is recorded (Diaz-Romero, Doshi, Hunn).
5. Reverse rule is universal and asymmetric: any neurological decline (Foster, Akins) — in Kuppler only if CT confirms re-accumulation — sends the drain to "a more negative pressure" and restarts the stepwise ascent. How far down is never stated (neither "back to the nadir" nor "one step back"), and no source describes a "hold at the current height" tier.
6. Refractory course is defined by repeated failure of the gate, and the sources then change the end-point rather than the gate: shunt from a negative height (Akins -3), ETV (Foster after 28 d-6.5 mo; Kuppler), lengthen the interval (Kuppler 3 → 7 d), or abandon the EVD for reservoir pumping/antigravity shunt at zero (Kuppler, Godoy).

D. Comparison across sources

Item	What the sources say	Agreement
Reference level	External auditory canal/meatus (Pang via Akins, Owler, Godoy, Kuppler, Lin, Wu's measurement); tragus (Doshi); midbrain (Foster).	Near-consensus on EAC; Foster is the outlier.
Prove failure at ≥ 0 first	Akins (+5...+10 then 0...+5), Kuppler (≥ 0 failed), Weisbrod (responded at 0), Salam (ICP < 5 with limited drainage).	Consistent: descent begins only after zero has failed.
Starting depth	Stated as a starting setting: 0 to -5 below EAM (Owler, 5 pts); 5-10 cm below midbrain (Foster, series); -5 (Diaz-Romero, 1 case); -10 below EAC (Kuppler, 1 case); -8 (Salam, 1 case, first recorded setting). Akins give only the range at which response occurred (-3 to -10), not a start. Lin's -5 to -20 is an uncited secondary summary.	Primary sources start at 0 to -10 cm H₂O ; no primary source starts below -10.
Reported floor	-8 (Galbarriatu, abstract); -10 = upper end of Akins' "typically -3 to -10" response range (no per-patient nadir given); -12 (Foster, one illustrated case; series nadir not reported); -15 cm below EAC (Godoy, upper end of the 5-15 cm range, two cases); Pang mean nadir -5.7 ± 3.6 mm Hg (≈ -8 cm H ₂ O, our conversion; individual nadirs not in the abstract); -30 cm H ₂ O (Bergsneider 1999 "as low as", 4 pts; Salam, refractory single case — partial response, ventricles "still ... larger than his prehospital baseline").	Reported settings mostly bottom out at -10 to -15; -30 appears only in Bergsneider's shunt-nonresponsive series and Salam's refractory case.
Descent step / interval	Doshi: 5 cm steps; Salam: ± 2 cm hourly to keep flow; Godoy: "progressively"; Akins, Foster, Owler, Hunn, Lin: not stated .	No source gives a fixed timetable for going down; the two that give any rule are single cases.
Target while down	Prescribed hourly flow: 5-15 mL/h (Hunn, 8 pts, abstract; Kuppler, 1 case), 5-10 (Diaz-Romero, 1 case), 10-15 (Salam's and Lin's literature summaries), 10-20 (Salam's own case, hour by hour). Daily volume as a prescribed component: > 200 mL/day (Pandey, abstract). Daily volume as an observed requirement, not a target: ~300 mL (Godoy, "daily debts around 300 mL"), 160 mL/day "needed to maintain a stable ventricle size" (Kuppler; the 180 mL/day figure is from the later reservoir-pumping phase, not the EVD). Output rising above the pre-LPH baseline is an observed signature, not a target (Akins; Foster "We, too, noted...").	Consensus that flow, not pressure, is titrated ; prescribed target varies 5-20 mL/h; daily volumes are mostly observations.
ICP monitoring	Via the EVD column (Wu: supine, flat, 15 min; Akins; Salam hourly at the start); intraparenchymal sensor in a minority (Godoy, Pandey). Hunn and Diaz-Romero use ICP turning positive as the recovery marker.	Universal that ICP is recorded; only Godoy/Pandey used a separate parenchymal sensor.
Clinical marker	Level of consciousness/GCS in every source; Kuppler adds speech and nausea/vomiting; Bruk clamp-test (GCS 15 open vs 13 clamped); Wang qEEG delta power.	Consensus on consciousness; qEEG is a single report.

Item	What the sources say	Agreement
Imaging cadence	CT before every upward step (Foster; implied by Akins' "radiographic stability"); "serial" CT (Kuppler, Godoy, Bruk); CT after each strategy change (Salam); Lin: one CT/MRI to set the baseline once ventricles are small. No source states a fixed interval such as 24-72 h.	Agreement on CT gating each ascent step; no agreed interval during the descent.
Stop-lowering rule	Ventricular size normalises / exam recovers (Pang, Bergsneider, Godoy, Doshi); output rises above baseline (Akins); flow target met (Hunn, Kuppler, Salam).	Implicitly the same rule everywhere: stop going down when CSF flows and ventricles/exam respond; there is no pressure floor defined as "enough".
Ascent rule	Akins 3 cm every 3-5 d; Foster 1-2 cm every 3-5 d; Diaz-Romero 1 cm every 3 d; Kuppler 2 cm every 3 d; Lin 1-2 cm every 3-5 d; Doshi 5 cm/day. Gate = clinical + CT stability; relapse → lower to "a more negative pressure" (depth unspecified) and restart the ascent (Akins, Foster, Kuppler).	Strong agreement on 1-3 cm every 3-5 days ; Doshi is the outlier.
End of EVD phase	Stable at 0 to +3 cm (Akins), ≥ 0 or +4 cm (Lin), at/above midbrain (Foster, +5 in the case), positive ICP (Diaz-Romero, Hunn), "Monro level" (Godoy), ventricular pressure > 50-60 mm H ₂ O plus sterile CSF/ no pneumocephalus (Wu).	Agreement that a shunt is placed once ventricles stay small at a neutral-to-slightly-positive height ; Wu adds infection preconditions.
Duration	Pang 22 d mean; Godoy 10-40 d (one case 25 d to normal size, shunt day 40); Foster 73 d mean without ETV (14-257); Chen 42 d (2-365); Kuppler 180 d unsuccessful.	Weeks to months; refractory cases defined by inability to climb.

Bottom line

1. Every series lowers the drain only after drainage at 0 to +10 cm has demonstrably failed; the documented opening settings run from 0 to -5 cm (Owler) through -5 (Diaz-Romero), -8 (Salam) and -10 (Kuppler) below the external auditory canal, or 5-10 cm below the midbrain (Foster). Akins report only the range at which response occurred (-3 to -10).
2. Descent is governed by CSF flow, not by a schedule: the drain is dropped until it drains (5-15 mL/h or output above the patient's pre-LPH baseline) and the ventricles shrink; there is no published interval or step size for the descent apart from two single cases (5 cm steps; ±2 cm hourly).
3. What is measured is consistent — level of consciousness, ICP through the drain, hourly/daily output and CT ventricular size — but only Foster ties CT to a rule (a scan before every upward step), and no paper fixes a reassessment interval such as 24-72 h.
4. "Stop lowering" = flow restored + ventricles smaller + exam better; pressure itself is not the target and no author names a pressure floor. Settings below -15 cm H₂O occur only in refractory cases (Bergsneider, Salam) and gave only a partial response in Salam.
5. The ascent is the standardised part: 1-3 cm every 3-5 days, each step gated by clinical and CT stability, with relapse sending the drain back to an unspecified more-negative height and restarting the ascent; shunt once stable at 0 to +5 cm (or positive ICP).

Part III — Paper listings by axis (computed from `index.tsv`)

✓FT = open-access full text mirrored under `fulltext/`. Evidence class in brackets; `cited` = Europe PMC citation count at harvest.

Core entity — low-/negative-pressure hydrocephalus

79 records

- **1974** McCullough DC et al. — Negative intracranial pressure hydrocephalus in adults with shunts and its relationship to the production of subdural hematoma. Journal of neurosurgery. [clinical / mechanistic study (adult), cited 77] [PMID 4813717](#)

- **1986** McCullough DC. — Symptomatic progressive ventriculomegaly in hydrocephalics with patent shunts and antisiphon devices. *Neurosurgery*. [case series (mixed), cited 37] [PMID 3491341](#)
- **1988** Foltz EL et al. — Symptomatic low intracranial pressure in shunted hydrocephalus. *Journal of neurosurgery*. [clinical / mechanistic study, cited 54] [PMID 3343612](#)
- **1993** Drake JM et al. — Functional obstruction of an antisiphon device by raised tissue capsule pressure. *Neurosurgery*. [case report (pediatric), cited 22] [PMID 8421544](#)
- **1993** Pomeranz S et al. — The effect of intracranial hypotension on cerebral blood flow in a feline model. *Acta neurochirurgica*. [experimental (animal / bench), cited 5] [PMID 8333302](#)
- **1994** Foltz EL et al. — Hydrocephalus: the zero ICP ventricle shunt (ZIPS) to control gravity shunt flow. A clinical study in 56 patients. *Child's nervous system : ChNS : official journal of the International Society for Pediatric Neurosurgery*. [case series / cohort, cited 4] [PMID 8194062](#)
- **1994** Pang D et al. — Low-pressure hydrocephalic state and viscoelastic alterations in the brain. *Neurosurgery*. [case series / cohort, cited 96] [PMID 7808607](#)
- **1995** Vassilyadi M et al. — Negative-pressure hydrocephalus. *Journal of neurosurgery*. [case report (pediatric), cited 32] [PMID 7666227](#)
- **1999** Bergsneider M et al. — Beneficial effect of siphoning in treatment of adult hydrocephalus. *Archives of neurology*. [case series / cohort (adult), cited 33] [PMID 10520938](#)
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- **2002** Kuba H et al. — Gait disturbance in patients with low pressure hydrocephalus. *Journal of clinical neuroscience : official journal of the Neurosurgical Society of Australasia*. [case series / cohort, cited 15] [PMID 11749014](#)
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- **2004** Gnanalingham KK et al. — 14 mmHg: a case of raised intracranial pressure?. *British journal of neurosurgery*. [case report (adult), cited 1] [PMID 15327239](#)
- **2006** Bergsneider M et al. — What we don't (but should) know about hydrocephalus. *Journal of neurosurgery*. [narrative review (mixed), cited 31] [PMID 16572631](#)
- **2006** Clarke MJ et al. — Very low pressure hydrocephalus. Report of two cases. *Journal of neurosurgery*. [case report, cited 17] [PMID 16961147](#)
- **2008** Rekate HL et al. — The importance of the cortical subarachnoid space in understanding hydrocephalus. *Journal of neurosurgery. Pediatrics*. [case report, cited 75] [PMID 18590387](#)
- **2008** Solheim O et al. — Rapid and severe rise in static and pulsatile intracranial pressures during a generalized epileptic seizure. *Seizure*. [case report, cited 20] [PMID 18554932](#)
- **2009** Garg A et al. — Endoscopic third ventriculostomy: 5 years' experience at the All India Institute of Medical Sciences. *Pediatric neurosurgery*. [case series / cohort (pediatric), cited 16] [PMID 19221456](#)
- **2011** Akins PT et al. — The genesis of low pressure hydrocephalus. *Neurocritical care*. [case report, cited 37] [PMID 21523524](#)
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- **2013** Preuß M et al. — Acute two-compartment low pressure hydrocephalus--a case report. *Child's nervous system : ChNS : official journal of the International Society for Pediatric Neurosurgery*. [case report (pediatric), cited 6] [PMID 23715811](#)
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- **2017** Barami K. — Letter: Low-Pressure Hydrocephalus in Children: A Case Series and Review of the Literature. *Neurosurgery*. [comment / letter (pediatric), cited 2] [PMID 28973300](#)
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Mechanism — brain biomechanics, compliance, elastography, modelling

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Treatment techniques — EVD titration, valves, drainage strategy

30 records

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Aetiological settings — trauma, craniectomy, SAH, tumour, infection

14 records

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Historical usage — 'low-pressure hydrocephalus' as a synonym for NPH

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